

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

NEURENT MEDICAL INC. & NEURENT MEDICAL LTD.,
Petitioner,

v.

THE FOUNDRY, LLC,
Patent Owner.

IPR2024-00279
Patent 10,610,675 B2

Before SHERIDAN K. SNEDDEN, SUSAN L. C. MITCHELL, and
TIMOTHY G. MAJORS, *Administrative Patent Judges*.

SNEDDEN, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)
Dismissing Patent Owner's Motion to Exclude
37 C.F.R. § 42.64(c)

I. INTRODUCTION

Neurent Medical Inc. and Neurent Medical Ltd. (collectively, “Petitioner”) filed a Petition requesting an *inter partes* review of claims 1–3, 7, 9, 10, and 17–25 of U.S. Patent No. 10,610,675 B2 (“the ’675 patent,” Ex. 1001). Paper 3 (“Pet.”). The Foundry, LLC (“Patent Owner”) filed a Preliminary Response to the Petition. Paper 7 (“Prelim. Resp.”).

We instituted trial on July 23, 2024. Paper 9 (“Inst. Dec.”). During trial, Patent Owner filed a Patent Owner Response. Paper 22 (“PO Resp.”). Petitioner filed a Reply (Paper 26 (“Pet. Reply”)) and Patent Owner filed a Sur-reply (Paper 30 (“PO Sur-reply”)). We held an oral hearing on April 29, 2025, and a transcript of that hearing is of record. Paper 40 (“Tr.”).

Patent Owner’s Motion to Exclude (Paper 35) remains pending. We address that motion in Section IV below.

We have jurisdiction under 35 U.S.C. § 6(b). After considering the full record developed through trial, we determine that Petitioner has proved by a preponderance of the evidence that the challenged claims are unpatentable. *See* 35 U.S.C. § 316(e). Our reasoning is explained below, and we issue this Final Written Decision under 35 U.S.C. § 318(a).

II. BACKGROUND

A. *Real Parties in Interest*

Neurent Medical Inc. and Neurent Medical Ltd. identify themselves as the real parties in interest for Petitioner. Pet. 3. Patent Owner identifies The Foundry, LLC as the owner of the ’675 patent and identifies Aerin Medical Inc. as the exclusive licensee of the ’675 patent. Paper 6, 2

B. Related Matters

The parties indicate that the '675 patent is involved in *Aerin Medical Inc. and The Foundry LLC v. Neurent Medical Inc. and Neurent Medical Ltd.*, Case No. 1:23-cv-00756-JLH (D. Del), filed July 11, 2023. Pet. 3; Paper 6, 2.

Petitioner has also filed petitions for *inter partes* review of the following patents asserted against Petitioner by Patent Owner and/or Aerin Medical Inc.: IPR2024-00275 against U.S. Patent No. 9,072,597; IPR2024-00276 against U.S. Patent No. 9,415,194; IPR2024-00277 against U.S. Patent No. 11,241,271; IPR2024-00278 against U.S. Patent No. 11,033,318; IPR2024-00280 against U.S. Patent No. 10,894,011; IPR2024-00281 against U.S. Patent No. 11,679,077; and IPR2024-00282 against U.S. Patent No. 11,766,286. Pet. 3.¹

C. The '675 patent (Ex. 1001)

The '675 patent is titled “Systems and Methods for Delivery of a Therapeutic Agent.” Ex. 1001, code (54). According to the '675 patent, “certain neurotoxins have found widespread use in millions of patients worldwide for a variety of conditions.” *Id.* at 1:43–47. The '675 patent states that neurotoxins have been proposed for controlling rhinorrhea and many other conditions. *Id.* at 2:17–46. The '675 patent explains that “[r]hinorrhea is the term describing the effluence of mucus from the lining of the nasal passages, nasopharynx, or paranasal sinuses” and “can be a

¹ Petitioner’s challenge in IPR2024-00281 was refiled and is now case IPR2024-00669. IPR2024-00281 has since been terminated. IPR2024-00281, Paper 10.

symptom of a number of diseases such as the common cold, sinusitis or rhinitis.” *Id.* at 7:37–41.

The ’675 patent states that “[d]ue to their extreme toxicity, neurotoxins are highly controlled and can have disastrous consequences if not used and controlled properly, especially when used in vivo.” Ex. 1001, 2:52–54. Thus, the ’675 patent explains that

it would be desirable to provide improved methods and systems for delivering toxins, such as botulinum and active botulinum fragments, to the nasal membrane of a patient, particularly a patient suffering from rhinitis or other conditions associated with nasal inflammation and conditions, such as sinus headaches and migraine headaches. The methods and systems should be capable of providing for selective and repeatable delivery of the toxins to defined target areas within the nasal cavities, including particular paranasal sinuses, the nasopharynx, and in some cases substantially the entire nasal cavity. The systems and methods should provide for the safe and effective delivery of the toxins, and in particular should reduce or eliminate the risk of toxin being delivered to non-targeted tissues outside of the nasal cavity.

Id. at 3:50–64.

The ’675 patent discloses “methods and apparatus for the control of autonomic nerve function comprising delivery via permeabilization of targeted cell membranes a therapeutically effective amount of a portion or fragment of a neurotoxin”, or “[m]ore particularly, the present invention relates to methods and systems for delivering toxins, such as botulinum toxin light chain fragments, to target cells in a nasal cavity.” Ex. 1001, 1:33–42. The ’675 patent states that

[a] further aspect of the invention is to provide methods and apparatus for treating or inhibiting a variety of diseases or syndromes that have an underlying neurogenic component, by disrupting the neurogenic activities of the promoters or

mediators of the disease or syndrome. Such disruption is facilitated by delivering an active fragment of botulinum toxin, such as the light chain portion of botulinum toxin serotype A (BoNT-LC/A), in the presence of an electric field or ultrasonic energy applied to permeabilize the wall of targeted cell under conditions which induce reversible poration of the cellular membrane and efficient passage of the BoNT-LC fragment to the cell cytosol, its catalytic environment.

Id. at 4:59–5:4.

The '675 patent describes delivering neurotoxins and active fragments thereof “to a patient’s nasal membrane while applying energy to target cells within the membrane under conditions which cause a reversible (or in some instances non-reversible) poration of the cell membranes to enhance delivery of the toxin into the cells.” Ex. 1001, 8:4–11. According to the '675 patent, the toxin may be introduced to “any portion of the nasal cavity, such as a single paranasal sinus or portion thereof, a main nasal passage, two or more paranasal sinuses,” “the entire nasal cavity of the patient,” or the “nasopharynx which is at the back of the nasal passage” and “comprises a cluster of goblet cells which are responsible for mucus secretion and which are susceptible to the disabling mechanism of the botulinum toxin and other neurotoxins.” *Id.* at 8:11–18. The '675 patent reports that targeting the nasopharynx allows “a high percentage of mucus-secreting epithelial or goblet cells [to] be treated with a device which is relatively small and which may carry a relatively low infusion volume and require less energy.” *Id.* at 24:2–6.

Further, the '675 patent describes:

The energy is preferably selectively applied to a targeted region containing a variety of cell types, including goblet cells, epithelial cells, ciliated and non-ciliated columnar cells, basal cells, and less or no energy applied to untargeted regions. It will

be appreciated that the energy may be applied to regions of the nasal membrane which are the same or different from the regions to which the toxin has been introduced. By controlling the delivery area of both the toxin delivery and the energy delivery, the methods and apparatus of the present invention can more specifically target the epithelial or goblet and other recipient cells of interest while minimizing the amount of toxin which enters non-targeted cells. That is, only those cells in the nasal membrane which are exposed to both the toxin and the applied energy will preferentially be permeablized or porated to receive the toxin within the cytoplasm of the cell.

Id. at 8:22–37; *see also id.* at 9:58–61 (“When the energy applicator is incorporated with or within the catheter, the delivery pattern will usually be at least partially overlapping with the toxin delivery pattern of the catheter.”). According to the patent, in the case where the regions of applied energy and toxin dispersion only partially overlap, “the delivery of the toxin will be enabled or enhanced principally within the regions of overlap”). *Id.* at 10:1–3.

The '675 patent describes systems which comprise “catheter[s] or other structure[s] adapted to introduce the toxin or toxin fragment to a region adjacent the target cells,” “an energy applicator for applying energy to the target cells under condition[s] which cause poration of the cell membranes to enhance delivery of the toxins and/or their active fragments,” and “a source of the toxin or active fragments suitable for introduction from the catheter or other delivery structure.” Ex. 1001, 5:11–19. According to the '675 patent, “[t]he energy applicator will typically be adapted to selectively apply energy to target cells within the region where the toxin is to be introduced, e.g., by focusing energy distribution to the particular target cells or regions which are rich with the target cells,” but “may be adapted to apply energy non-selectively within the target region where both target cells and other cell

types may receive the energy.” *Id.* at 5:20–27. The energy applicator may apply electric energy (such as radiofrequency energy, direct current, alternating current, or combinations thereof), ultrasonic energy (such as high acoustic pressure ultrasound or low acoustic pressure ultrasound), X-ray beam energy, microwave energy, or “any other energy type which can achieve a reversible poration of the target cell walls.” *Id.* at 5:35–53.

D. The Challenged Claims

Petitioner challenges claims 1–3, 7, 9, 10, and 17–25. Representative independent claims 1 and 17 are reproduced below.

1. A method of treating a nasal condition of a patient comprising:

positioning a distal portion of a treatment device proximate to target tissue in a nasopharyngeal passage of a patient; and

administering therapy to the target tissue with the treatment device so as to disrupt neural activity such that mucus secretion is reduced in the nasopharyngeal passage;

wherein administering therapy comprises at least one of delivering energy to the target tissue and changing the temperature of the target tissue.

17. A method of treating a nasal condition of a patient comprising:

positioning a distal portion of a treatment device proximate to target tissue in a nasopharyngeal passage of a patient; and

administering therapy to the target tissue with the treatment device to alter nasopharyngeal cells such that congestion in the nasopharyngeal passage is reduced;

wherein administering therapy comprises delivering energy to the target tissue.

Ex. 1001, 27:39–50, 28:22–31.

E. Evidence

Petitioner relies upon information that includes the following.

Ex. 1003, Saadat, US 10,307,200 B2, issued June 4, 2019 (“Saadat”).

Ex. 1004, Deem et al., US 2008/0021369 A1, published January 24, 2008 (“Deem”).

Ex. 1005, Makower et al., US 2005/0240147 A1, published October 27, 2005 (“Makower”).

Ex. 1006, Fang and Zhen, *Nasal endoscopy combined with multiple radiofrequency for perennial allergic rhinitis*, 25(7) J. FIRST MIL. MED. UNIV. 876–77 (2005) (“Fang”).

Petitioner also relies on declaration testimony from Dr. Nicolas Bu-Saba (Exs. 1002 and 1098) in support of its challenges. Patent Owner relies on declaration testimony from Dr. Howard L. Levine (Ex. 2003) and Dr. Marom Bikson (Ex. 2005).

Deposition testimony of the above declarants is also of record. Exs. 2002 and 2009 (Bu-Saba transcripts); Ex. 1089 (Levine transcript); Ex. 1091 (Bikson transcript).

F. Asserted Grounds of Unpatentability

Petitioner asserts that claims 1–3, 7, 9, 10, and 17–25 would have been unpatentable on the following grounds:

Ground	Claim(s) Challenged	35 U.S.C. §	Reference(s)/Basis
1	1–3, 7, 9, 10, 17–25	102	Saadat
2	24, 25	103	Saadat
3	1–3, 7, 9, 10, 17–25	102	Deem
4	1–3, 7, 9, 10, 17–25	103	Makower, Fang

III. ANALYSIS

A. Legal Standards

“In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable.” *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3)).

A claim is unpatentable under 35 U.S.C. § 103 if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the relevant art. *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) secondary considerations (i.e., objective indicia) of nonobviousness, when presented.² *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966). “An obviousness determination [also] requires finding both that a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.” *CRFD Rsch., Inc. v. Matal*, 876 F.3d 1330, 1340 (Fed. Cir. 2017) (internal quotation marks and citation omitted).

² The parties did not present argument or evidence in this case related to objective indicia of nonobviousness.

B. Level of Ordinary Skill in the Art

In determining the level of ordinary skill in the art, various factors may be considered, including the “type of problems encountered in the art; prior art solutions to those problems; rapidity with which innovations are made; sophistication of the technology; and educational level of active workers in the field.” *In re GPAC, Inc.*, 57 F.3d 1573, 1579 (Fed. Cir. 1995). The prior art itself can also reflect the appropriate level of ordinary skill in the art. *Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001).

Petitioner proposes that the person of ordinary skill in the art (“POSA”) at all relevant times (between the years 2005 through 2019) would have had:

(1) a Doctor of Medicine (M.D.) or at least a bachelor of science degree in engineering or a related discipline and (2) at least three years of experience in the clinical use, design, development, and/or testing of medical devices, including medical devices that deliver energy to tissue. Less work experience may be compensated by a higher level of education, and vice versa.

Pet. 17 (citing Ex. 1002 ¶¶ 26–30).

Patent Owner disagrees with Petitioner’s definition of the POSA’s qualifications. PO Resp. 18–20. Patent Owner contends that “Petitioners’ definition allows for a person with no knowledge of nasal anatomy to opine on issues in this proceeding.” *Id.* at 18. Patent Owner contends that a POSA “must have knowledge of the nasopharyngeal passage given the scope of the claims.” *Id.* at 18–19. Patent Owner further contends that Petitioner’s definition is “improper” because, under that definition, the POSA could be an M.D. with clinical experience using relevant medical devices, but no specific “medical device design or development experience.” *Id.* at 19 (citing Ex. 2003 ¶ 37).

Patent Owner alternatively proposes that a POSA should be defined as a person having:

- (1) a Doctor of Medicine (M.D.) or at least a Bachelor of Science degree in engineering or a related discipline and
- (2) at least three years of experience in the design and development of nasal medical devices, including nasal medical devices that deliver energy to tissue.

Id. at 19 (citing Ex. 2003 ¶ 38) (emphasis omitted).

Finally, Patent Owner contends that the testimony of Petitioner's expert should be given "little weight" because Dr. Bu-Saba does not have device design/development experience. PO Resp. 19 n.4 (citing Ex. 2002, 43:20–45:11).

In response, Petitioner contends that Dr. Bu-Saba qualifies as a POSA and that his testimony should be credited on the issues in dispute in this case. Pet. Reply 27–29.

Petitioner argues, for example, that Dr. Bu-Saba is a clinician, nasal cavity/nasopharynx anatomy expert, and that his "testimony is supported by his extensive clinical expertise." Pet. Reply 27 (citing Ex. 1089 (Levine Tr.), 170:21–171:2 (recognizing Dr. Bu-Saba's experience with RF ablation procedures and conceding Dr. Bu-Saba is an expert on the anatomy of the nasal cavity and nasopharynx)). Petitioner also argues that Patent Owner's definition is flawed insofar as it limits the POSA to one having design or development experience, which Petitioner contends is not required. *Id.* at 28–29. In support, Petitioner argues, *inter alia*, that: 1) the challenged claims relate to delivering RF energy with generic (not specialized) components (*id.* at 28 (citing Ex. 1001, 27:43–28:47)); 2) "[c]linicians used RF devices safely and effectively in the nose, including using 'factory settings'" (*id.* (citing Ex. 1001, 16:42–49, 18:5–10, 18:24–36)); 3) clinicians

understood and carried out safe and effective RF device operation through, e.g., clinical use and testing (*id.* at 29 (citing Ex. 1089, 62:7–63:11 (Dr. Levine admitting safe and effective operation is learned through clinical use and instructions))); and 4) clinicians need not have developed a device to “see another place it could be used” (*id.* (quoting Ex. 1089, 124:19–125:13))).

To further support its contentions, Petitioner points out, *inter alia*, that: 1) “[t]he ’675 patent discusses ‘delivering neurotoxins’ in ‘*any* . . . target region,’” not just the nose, and “identifies no purportedly challenging nasal anatomy” (Pet. Reply 30 (citing, e.g., Ex. 1001, 2:23–67, 5:3–17, 4:59–63, 8:10–34, claim 1)); 2) “[p]rior-art problems and solutions were addressed with ‘factory setting’ generic RF devices” (e.g., Ex. 1006, 876–77), and “devices for multiple locations” (e.g., Ex. 1005 ¶ 43 (ear, nose, throat); Ex. 1009, 1:15-23) (Pet. Reply 32); 3) “[t]he named inventors are engineers, and [Patent Owner] offers no evidence either had prior nasal experience” to support its position (*id.* at 31); and 4) “Dr. Levine confirmed biomedical engineers ‘take in information from clinicians’ to create devices” (*id.* (citing Ex. 1089, 202:6–203:2 (Dr. Levine agreeing that “it’s the job of a biomedical engineer, working on a medical device, to take in information from clinicians to create better tools.”)))).

In its Sur-Reply, Patent Owner reiterates its disagreement with Petitioner’s POSA definition, arguing that definition would allow “an engineer without any RF or nose experience to assess and apply the prior art.” PO Sur-Reply 25.

Based on the record before us, including the types of problems and solutions described in the ’675 patent and the cited prior art, we find a POSA would have had (1) a Doctor of Medicine (M.D.) or at least a bachelor of

science degree in biomedical engineering or a related discipline, (2) at least three years of experience in the clinical use, design, development, and/or testing of medical devices that deliver energy to tissue, and 3) knowledge and experience in treating nasal conditions, or access to a person with that knowledge and experience. Less work experience may be compensated by a higher level of education, and vice versa.

In reaching our finding, we note that, although we agree with Petitioner that the '675 patent discloses delivering neurotoxins to any target, and not just the nose, the claims of the '675 patent are directed to methods of treating a nasal condition by delivering energy to the nasal cavity. Pet. Reply 30. Thus, to fill a gap between Petitioner's POSA definition and the scope of the challenged claims, we add to Petitioner's definition the requirement for knowledge and experience in treating nasal conditions, or access to a person with that knowledge and experience. We additionally add that a POSA's educational background may include a bachelor of science degree in biomedical engineering or related discipline to reflect the nature of the field of invention. Nonetheless, we also note that our analysis and conclusions herein would not change even under Petitioner's proposed skill level for a POSA.

We do not agree with Patent Owner that prior experience designing or developing nasal medical devices is required for a POSA. PO Resp. 19. A definition inclusive of the clinical use, design, development, and/or testing of nasal medical devices that deliver energy to tissue is consistent with the skill level reflected in the disclosures of the '675 patent and prior art. Consequently, we disagree with Patent Owner that Dr. Bu-Saba fails to qualify as a POSA because he lacks experience specific to *designing* medical devices. PO Resp. 19 n.4. Accordingly, we do not agree with Patent Owner

that we should discount the testimony of Dr. Bu-Saba and we give Dr. Bu-Saba's testimony appropriate weight in reaching our conclusions here.

C. Claim Construction

In *inter partes* review, we construe claims using the same claim construction standard used to construe claims in a civil action before the courts under 35 U.S.C. § 282(b), including construing claims' language in accordance with its ordinary and customary meaning as understood by the POSA, in view of the patent's specification and considering the patent's prosecution history. 37 C.F.R. § 42.100(b). We need only construe terms that are in controversy and only to the extent needed to resolve the dispute. *Realtime Data, LLC v. Iancu*, 912 F.3d 1368, 1375 (Fed. Cir. 2019); *U.S. Surgical Corp. v. Ethicon, Inc.*, 103 F.3d 1554, 1568 (Fed. Cir. 1997) (holding claim construction is for "resolution of disputed meanings").

Petitioner contends that, "[f]or purposes of this IPR, the Board need not construe any terms." Pet. 18. Patent Owner responds that "the Board need not construe any terms to determine Petitioners have failed to carry their burden in this proceeding." PO Resp. 20.

On the record before us, we conclude that the claims need not be expressly construed to resolve the dispute before us. Insofar as the parties dispute the mapping of the prior art to dependent claim 22, we address that dispute below including, to the extent necessary, any implicit disagreement about the subject matter encompassed by claim 22.

D. Effective Filing Date of the Challenged Claims

The parties dispute whether the challenged claims are entitled to priority earlier than October 22, 2019, and relatedly, whether Saadat and Deem qualify as prior art. Pet. 18–26 (arguing, *inter alia*, breaks in the

priority chain among the ancestral applications to which the '675 patent purports to claim priority, and the challenged claims lack written description support in any application filed before October 22, 2019); PO Resp. 20–31 (arguing the patent includes an unbroken priority claim to July 21, 2006, and the intervening applications provide descriptive support for the claims).

The '675 patent includes a long and somewhat complex priority chain. *See* Inst. Dec. 10–11 (discussing and depicting the asserted priority chain). And the parties' priority arguments involve hotly-contested legal and factual issues. *See, e.g.*, Pet. 24–25 (e.g., whether Patent Owner can rely on a non-provisional application's "nested" incorporation of subject matter from a provisional application, contrary to 37 C.F.R. § 1.57(d)); PO Resp. 23–25 (arguing the priority applications show possession "of the use of energy alone" to treat nasal conditions and reduce congestion); Pet. Reply 3–8 (arguing that, although the claims "recite energy-only treatment," the priority applications "do[] not describe *treatment* absent toxin"—instead they provide written description support only for an apparatus/method that provides treatment through *combined delivery of energy and toxin*)).

We need not decide these disputed priority issues to resolve the controversy in this case. Even if there were no insurmountable legal or factual barrier to the claims receiving a priority date of July 21, 2006 (as argued by Patent Owner), that would still leave Ground 4. Pet. Reply 10 ("Ground IV renders the Challenged Claims obvious at any priority date"). And, as we explain below, we determine that the preponderance of the evidence shows that all the challenged claims are unpatentable under Ground 4.

E. Asserted References

Saadat and Deem form the basis of Petitioner’s assertions in Grounds 1–3. Whether Saadat or Deem qualify as prior art turns on whether the challenged claims are entitled to claim priority no earlier than October 22, 2019. But, because we do not decide that priority issue in this Decision and do not reach the Grounds that rely on Saadat or Deem (Grounds 1–3), we omit summaries of Saadat and Deem below.

As it concerns Ground 4, there is no dispute that Makower and Fang are prior art against the challenged claims. Those references are summarized below.

1. Makower (Ex. 1005)

Makower is a U.S. patent application that published October 27, 2005. Ex. 1005, code (43).

Makower is titled “Devices, Systems and Methods for Diagnosing and Treating Sinusitis [*sic*] and Other Disorders of the Ears, Nose and/or Throat.” *Id.* at code (54). Makower “relates generally to medical devices and methods and more particularly to minimally invasive, catheter based devices, systems and methods for treating sinusitis and other ear, nose & throat disorders.” *Id.* ¶ 1. Makower states that “[t]he term ‘sinusitis’ refers generally to any inflammation or infection of the paranasal sinuses.” *Id.* ¶ 8.

Makower explains that “[t]he sinuses consist of a series of cavities connected by passageways, ultimately opening into the nasal cavity.” Ex. 1005 ¶ 16. The “passageways and cavities are formed by bone, but covered in mucosa.” *Id.* Makower reports that “[i]f the mucosa of one of these passageways becomes inflamed for any reason, the cavities which drain through that passageway can become blocked” such that “the primary

target for sinus therapy is the passageway, with the primary goal to regain drainage.” *Id.* According to Makower, “focus on the passageway and the use of flexible instrumentation suggests an entirely new approach to sinus intervention: one utilizing flexible catheters and guidance tools, with passageway and cavity modifying devices capable of being delivered with minimal damage to the surrounding tissues.” *Id.*

One such intervention is surgery to reduce the size of the inferior turbinates and the associated mucosa or soft tissue. Makower explains that

[s]urgery to reduce the size of the inferior turbinates can be accomplished with endoscopic visualization (with magnification where desired) and is typically performed with the patient under general anesthesia. An incision is typically made in the mucosa that lines the turbinate to expose the underlying bone. Some quantity of the underlying bone may then be removed. If selective removal of some of the mucosa or soft tissue is also desired, such soft tissue can be debulked or removed through by traditional surgical cutting or by the use of other tissue ablation or debulking apparatus such as microdebriders or lasers. Less frequently, chronically enlarged inferior turbinates have been treated by cryotherapy. It is typically desirable to remove only as much tissue as necessary to restore normal breathing and drainage, as removal of too much tissue from the turbinates can impair the ability of the turbinates to perform their physiological functions of warming and humidifying inspired air and conserving warmth and moisture from expired air.

Ex. 1005 ¶ 30.

Another example of an intervention disclosed in Makower is laser or radiofrequency turbinate reduction. According to Makower, “[i]n cases where it is not necessary to revise the bone that underlies the turbinate, the surgeon may elect to perform a laser or radiofrequency procedure designed to create a coagulative lesion in (or on) the turbinate, which in turn causes the soft tissue of the turbinate to shrink.” *Id.* ¶ 39. Makower discloses that

“[o]ne example of a radio frequency procedure that may be used to shrink enlarged inferior turbinates is radiofrequency volumetric tissue reduction (RFVTR) using the Somnoplasty® system (Somnus Medical Technologies, Sunnyvale, Calif.)” which “includes a radio frequency generator attached to a probe.” *Id.* ¶ 40. Makower explains that

[t]he probe is inserted through the mucosa into the underlying soft tissue of the turbinate, usually under direct visualization. Radiofrequency energy is then delivered to heat the submucosal tissue around the probe, thereby creating a submucosal coagulative lesion while allowing the mucosa to remain intact. As the coagulative lesion heals, the submucosal tissue shrinks thereby reducing the overall size of the turbinate.

Id.

Makower provides “methods wherein one or more flexible catheters or other flexible elongate devices as described herein are inserted [into] the nose, nasopharynx, paranasal sinus, middle ear or associated anatomical passageways to perform an interventional or surgical procedure.” Ex. 1005

¶ 43. Makower discloses that its

invention provides a variety of apparatus that may be inserted into the nasal cavity, paranasal sinus, nasopharynx or middle ear to perform diagnostic or therapeutic procedures. These devices may be delivered through or incorporated into flexible catheters or flexible rod-like shafts. Such flexible construction allows these devices to be delivered and positioned to perform the desired diagnostic or therapeutic procedures with minimal trauma to other tissues, as can result from the insertion of rigid scopes and rigid instruments in accordance with the methodology of the prior art.

Id. ¶ 161.

An example of a tissue shrinking electrode device being used in accordance with Makower’s invention is shown in Figure 5H, reproduced below.

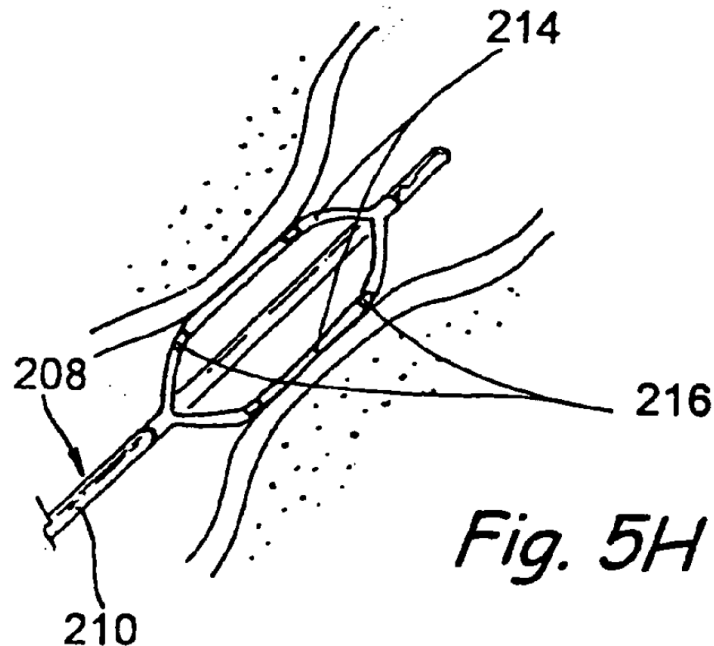


Figure 5H depicts “an electrosurgical device 208 comprising a flexible shaft 210 (e.g., a catheter or solid shaft) having arched strut members 214 attached thereto.” Ex. 1005 ¶ 187 (emphasis omitted). The strut members 214 may be of fixed configuration or collapsible and expandable, and include electrodes 216 that “may be bipolar, monopolar or facilitated by any other suitable form of energy such as a gas or plasma arc.” *Id.* Makower discloses that “sensing elements may also be attached to the catheter and/or strut members to monitor various parameters of the catheter and/or surrounding tissue (e.g., temperature, etc.).” *Id.* According to Makower, “[i]n operation, the device 208 is advanced into a passageway such as a nostril, nasal cavity, meatus, ostium, interior of a sinus, etc. Thereafter, current is applied to the electrodes 216 causing tissue adjacent to the struts 214 to be cauterized or heated.” *Id.* (emphasis omitted).

2. *Fang (Ex. 1006)*

Fang is an article titled “Nasal endoscopy combined with multiple radiofrequency for perennial allergic rhinitis.” Ex. 1006, 876.³ Fang discusses the results of a study “[t]o evaluate the therapeutic effects of nasal endoscopic surgery combined with multiple radiofrequency in the treatment of perennial allergic rhinitis.” *Id.* According to Fang, “[s]ixty-three patients with perennial allergic rhinitis whose symptoms failed to be controlled by medication received nasal endoscopic surgery for submucous resection of the nasal septum combined with radiofrequency ablation of the bilateral sphenopalatine foramen, anterior ethmoidal nerve, and inferior nasal concha.” *Id.* “[A]nother 20 patients with the same disease who underwent nasal septum surgery alone were selected as the control group.” *Id.* After the surgeries, “efficacy was evaluated according to the evaluation criteria for allergic rhinitis” (*id.*), and “[a]ll the patients were followed up for 2 years, and no antihistamines or corticosteroids were used for treatment during the observation period” (*id.* at 877).

Fang reports that

[a]t 1 month after operation, the rate of excellent recovery was 92.1% (58/63), the efficacy rate was 7.9% (5/63), and the total efficacy rate was 100%. At 1 year after operation, the rate of excellent recovery was 88.9% (56/63), the efficacy rate was 7.9% (5/63), the rate of no efficacy was 3.2% (2/63), and the total efficacy rate was 96.8% (61/63). At 2 years after operation, the rate of excellent recovery was 82.6% (52/63), the efficacy rate was 12.7% (8/63), the rate of no efficacy was 4.8% (3/63), and

³ Fang is a Chinese to English translation of the original article; Petitioner submits a declaration (Ex. 1073) attesting to the translation’s accuracy, and testimony from June Munford (Ex. 1070 ¶¶ 39–56) on the article’s public availability by at least July 20, 2005. Patent Owner does not challenge the accuracy of this translation or the publication date of Fang.

the total efficacy rate was 95.2% (60/63). Compared with patients in the control group who underwent only nasal endoscopic surgery for submucous resection of the nasal septum, the total efficacy rate ($P < 0.01$ for excellent recovery + Efficacy) at 1 and 2 years after the surgery was significantly improved, with statistical significance.

Ex. 1006, 876.

According to Fang,

[t]he use of radiofrequency technique completely blocks the parasympathetic pathway of the nasal mucosa, reduces the vascular tension, reduces glandular secretion, and thus achieves the therapeutic purposes. Pterygoid canal neurotomy alone or/and anterior ethmoidal neurotomy or inferior turbinate treatment preserves 1-2 parasympathetic pathways in the source and distribution of parasympathetic nerves in the nasal mucosa. However, due to the presence of allergic reaction arcs, the postoperative effect is poor and the recurrence rate is high. For the patients in this group, all the nasal mucosal parasympathetic pathways, including the posterior ganglionic fibers of the sphenopalatine nerve, the parasympathetic fibers of the anterior ethmoidal nerve, and the cholinergic ganglia within the inferior turbinate mucosa were blocked. The allergic reflex arc was completely relieved, thus achieving stable and reliable results.

Id. at 877.

F. Ground 4: Asserted Obviousness over Makower and Fang

Petitioner contends that claims 1–3, 7, 9, 10, and 17–25 would have been obvious over the combination of Makower and Fang. Pet. 64–92. Petitioner reasons, *inter alia*, that a person of ordinary skill would have been motivated to combine the cited references “to apply Fang’s method (disrupting neural activity using ablation) with Makower’s device (electrosurgical device for the nasopharynx) to treat rhinitis.” *Id.* at 72.

Patent Owner argues there was no reason to combine the teachings of Makower and Fang, nor a reasonable expectation of success in doing so to arrive at the claimed subject matter. PO Resp. 31–65.

We discuss the parties’ arguments and evidence in more detail below, beginning with independent claim 1, then proceed to the remaining claims challenged under Ground 4.

1. Claim 1

a) Disclosure of Claim 1’s Limitations

Petitioner contends that the combination of Makower and Fang teaches or suggests all of claim 1’s limitations. Pet. 64–71. According to Petitioner, Makower and Fang together teach a method for treating rhinitis in a nasopharyngeal passage,⁴ meeting claim 1’s preamble (if that preamble language is limiting).⁵ *Id.* at 64–65. Petitioner cites, for example, Makower’s disclosure of “medical devices . . . for treating sinusitis . . . , [and other] nose & throat disorders.” Pet. 65 (citing Ex. 1005 ¶¶ 1, 43, 161, 187, claim 1; Ex. 1002 ¶¶ 151–152). Petitioner cites Makower’s disclosure of flexible catheters and working devices for insertion into the nose to perform interventional or surgical procedures (Ex. 1005 ¶¶ 43, 161), including, for example, electrosurgical treatment devices. Pet. 65–66 (citing Ex. 1005, Fig. 5H, ¶¶ 164, 187); Ex. 1002 ¶¶ 155–156). According to Petitioner, “Makower Figure 5H . . . teaches an electrosurgical device with electrodes on strut members 214 contacting surrounding tissue in the nasopharynx.” *Id.*

⁴ The nasopharyngeal passage includes the nasal cavity and nasopharynx. Ex. 1001, 23:66–24:6; *see also id.* Figures 18–19, 10:62–63; Pet. 33 n.15.

⁵ We need not decide whether the preamble language is limiting because there is no dispute that this subject matter is taught in the asserted art.

(citing Ex. 1005 ¶¶ 164 (devices may include “radiofrequency electrodes”), 187 (disclosing “current is applied to the electrodes 216 causing tissue adjacent to the struts 214 to be cauterized or heated”)).

Turning to Fang, Petitioner contends that Fang teaches “performing nasal endoscopic surgery combined with multiple radiofrequency” ablation to treat “perennial allergic rhinitis.” *Id.* at 66 (citing Ex. 1006, 876–877; Ex. 1002 ¶ 155). Petitioner cites Fang’s teaching that “radiofrequency ablation of [the] sphenopalatine foramen . . . block[s] the posterior parasympathetic fibers of the sphenopalatine nerve . . . [and] completely blocks the parasympathetic pathway of the nasal mucosa,” which “reduces glandular secretion, and thus achieves the therapeutic purposes.” Ex. 1006, 877; Pet. 66–67 (citing Ex. 1002 ¶¶ 155, 157); *see also* Pet. 67 (noting the sphenopalatine foramen (SPF) is located in the posterior nasal cavity)⁶; Ex. 1002 ¶ 158.

Petitioner contends that Makower and Fang disclose a treatment device with a distal portion configured for positioning in the nasopharyngeal passage as recited in claim 1. Pet. 68 (citing Ex. 1005 ¶ 161 (“Makower and Fang each teach “*positioning a distal portion of a treatment device proximate to tissue in the nasopharyngeal passage.*”); *see also* Ex. 1005 ¶¶ 133–135, Fig. 2A (occluder access device for positioning distal portion of flexible shaft in the nasal cavity); Ex. 1006, 876–877 (teaching a “radiofrequency therapy instrument” with distal “firing head” that a physician aligns with the SPF in nasal cavity); Ex. 1002 ¶¶ 160–164.

⁶ Doctors Bu-Saba and Levine provide further technical background about the relevant anatomy of the nasal cavity, including the SPF and nerves that control mucus producing cells and the relation of the same to rhinitis and its symptoms. *See, e.g.*, Ex. 1002 ¶¶ 31–33; Ex. 2003 ¶¶ 41–43.

Further, Petitioner contends, Makower teaches an electrosurgical device that may include multiple bipolar or monopolar electrodes that apply current to “tissue adjacent to the struts . . . to be cauterized or heated.” Pet. 68 (citing Ex. 1005 ¶ 187, Fig. 5H). As Dr. Bu-Saba explains, Makower teaches the well-known concept of treating tissue using bipolar electrodes to administer RF energy through electrical current. Ex. 1002 ¶¶ 166–167.

Claim 1 also recites “delivering energy to the target tissue and changing the temperature of the target tissue.” *See supra* Section II.D. As discussed above, Petitioner argues that Makower and Fang disclose delivery of RF energy to the nasal cavity via electrodes to ablate tissues to disrupt neural activity. Pet. 70–71 (citing Ex. 1002 ¶ 166). Petitioner contends that a POSA “would have understood that delivering RF energy to tissue, as taught by Makower and Fang, heats it and thereby changes its temperature.” *Id.* at 71 (citing Ex. 1002 ¶ 166).

Based on the argument and evidence summarized above, Petitioner persuades us that the combination of Makower and Fang teaches or suggests all of claim 1’s limitations. Patent Owner provides no argument to the contrary. *See* Pet. Reply 10 (“PO does not dispute Makower and Fang . . . teach all claim elements”). Patent Owner instead argues that Petitioner has set forth no adequate motivation to combine Makower and Fang and “the combination fails to establish a reasonable expectation of successfully achieving” the invention. PO Resp. 31 (citing Ex. 2003 ¶¶ 86–98; Ex. 2005 ¶¶ 57–72); *see generally id.* at 32–45 (counterargument on reasonable expectation of success), 45–63 (counterargument on motivation to combine); PO Sur-Reply 11–22. We address the disputed motivation and reasonable expectation of success issues below.

b) Motivation to Combine & Reasonable Expectation of Success

Petitioner contends that the skilled artisan would have been motivated to combine the teachings of Makower and Fang and done so with a reasonable expectation of success in arriving at the method of claim 1. Pet. 71–77; Pet. Reply 10–23. We summarize Petitioner’s position and evidence below.

Petitioner contends that Makower and Fang are analogous art to the challenged claims. Pet. 71 (arguing Makower and Fang “are in the same field” and are “reasonably pertinent” to solving problems addressed by the claimed invention); Ex. 1002 ¶ 169. According to Petitioner, Makower and Fang teach that their devices treat disorders of the nasal cavity by applying energy to tissue and “each uses RF energy to treat tissue.” Pet. 71–72 (citing, e.g., Ex. 1005 ¶¶ 1, 43, 132, 164, 187; Ex. 1006, 876–877).

Petitioner contends that a skilled artisan would have been motivated to use Makower’s electrosurgical device to apply RF ablation energy to regions of the nasal cavity to disrupt neural activity as disclosed in Fang. Pet. 72–73 (citing Ex. 1002 ¶¶ 170–171). According to Petitioner and its declarant Dr. Bu-Saba, a skilled artisan “would have understood from Fang that Makower’s electrodes, like the similar RF device in Fang, would treat nasal disorders by disrupting neural activity in the same way as Fang, by heating parasympathetic nerves downstream of the SPG and Vidian nerve, thereby inhibiting their signaling of mucus-producing goblet cells.” *Id.* at 72; Ex. 1002 ¶ 170 (testifying “Makower’s device would not require any modifications to achieve this result because Makower specifically teaches that its device applies energy to the nasal cavity and nasopharynx using electrodes, the same mechanism as Fang”).

According to Petitioner, a POSA “would have been motivated to combine Fang’s teachings regarding the known outcome of application of RF therapy with Makower’s device.” Pet. 75. Petitioner contends Makower discloses that, by positioning strut members and RF electrodes in the nasal cavity, RF energy may be applied to heat/cauterize nasal tissues adjacent the electrodes and treat nasal disorders. Pet. 75–76 (citing, e.g., Ex. 1005 ¶¶ 161, 164, 187, claim 1). Also, Petitioner contends, Fang explains that “[n]umerous studies [had] shown that the parasympathetic nerves in the nasal cavity . . . regulate the vascular dilation and contraction of the nasal mucus membrane and the secretion of glands.” *Id.* at 73 (citing Ex. 1006, 877). Thus, Petitioner argues, “Fang teaches the well-known fact that the parasympathetic nerves regulate mucus production”—providing a reason to target those nerves for ablation with Makower’s RF-electrosurgical device to reduce mucus secretions and treat rhinitis as taught in Fang (and as known elsewhere in the prior art). *Id.* at 73–74 (citing, e.g., Ex. 1006, 877; Ex. 1044,⁷ 240–241 (teaching RF thermocoagulation of parasympathetic nerves in the nasal cavity reduces symptoms of rhinitis)); Ex. 1051,⁸ 284;

⁷ Chen, et al., *Preliminary study on radiofrequency thermocoagulation of the posterior inferior nerve, anterior ethmoidal nerve, and inferior turbinate under nasal endoscopy for the treatment of perennial allergic rhinitis*, 11:3 CHINA JOURNAL OF ENDOSCOPY 239–243 (Mar. 2005) (Ex. 1044, “Chen”). Petitioner also provides as evidence an uncontested translator’s declaration for the Chen reference. Ex. 1071.

⁸ Robert E. Windsor, MD, and Scott Jahnke, DO, *Sphenopalatine Ganglion Blockade: A Review and Proposed Modification of the Transnasal Technique*, 7:2 PAIN PHYSICIAN 283–286 (2004) (Ex. 1051, “Windsor”). Windsor teaches, *inter alia*, that “[p]ost-ganglionic parasympathetic neurons from this ganglion [(the SPG)] then distribute to the lacrimal glands and glands of the nasal cavity . . . where they promote secretion”).

Ex. 1002 ¶¶ 174 (testifying it was “well-known to those skilled in the art that disrupting the Vidian nerve, SPG, or associated parasympathetic nerve branches would reduce the symptoms of rhinitis, i.e., mucus production, mucus secretion, and congestion” (citing Fang, Chen, and other studies in the technical literature as of 2005)).

Petitioner argues that the POSA would have reasonably expected success in combining the asserted prior art. Pet. 83–85. According to Petitioner, a POSA “would have reasonably expected Makower’s electrosurgical device to obtain Fang’s clinical result” because “Fang and others taught [that] applying RF energy to . . . sphenopalatine nerves treats rhinitis.” *Id.* at 76–77 (citing, e.g., Ex. 1044 (Chen), 239). As highlighted by Petitioner, “Chen explained ‘parasympathetic nerves in the nasal cavity mainly originate from the sphenopalatine nerves,’ and applying RF energy to them ‘block[s] the source of the parasympathetic nerves in the nasal cavity and reduce[s] the excitability of the parasympathetic nerves’ to disrupt rhinitis symptoms.” *Id.* at 76 (quoting Ex. 1044, 240–241); *see also* Ex. 1006 (Fang), 876–877 (concluding that “[n]asal endoscope combined with radio frequency [ablation of parasympathetic nerves in the nasal cavity] may achieve good therapeutic effects on perennial allergic rhinitis with less adverse effects”).⁹ Petitioner further contends that both Makower and Fang

⁹ As discussed in more detail above (Section III.E.2), Fang describes a study on 63 patients with perennial allergic rhinitis (PAR) who received a resection of the nasal septum “combined with radiofrequency ablation of the bilateral sphenopalatine foramen, anterior ethmoidal nerve, and inferior nasal concha” versus a control group of 20 PAR patients “who underwent nasal septum surgery alone.” Ex. 1006, 876. Fang reports that “[c]ompared with patients in the control group . . . , the total efficacy rate ($P < 0.01$ for

are directed to treating nasal conditions with RF energy-delivery devices, and thus, a POSA “would have had a reasonable expectation of successfully using Makower’s known RF device according to a known method to reach the predictable result taught in Fang and Chen.” Pet. 76–77 (citing Ex. 1002 ¶¶ 177–179).

We find persuasive Petitioner’s above argument on the motivation to combine and reasonable expectation of success issues, and we credit Petitioner’s evidence in support of the same. We address Patent Owner’s counterargument below and explain why it does not rebut Petitioner’s challenge to claim 1.

(1) *Alleged Lack of Reasonable Expectation of Success in Treating Nasal Tissue with the Makower Device*

Patent Owner contends that “Petitioners’ case is defective because Petitioners failed to establish that Makower’s FIG. 5H device would reasonably be expected to properly deliver energy to target tissue to achieve the claimed therapeutic results.” PO Resp. 33. More specifically, Patent Owner argues: (1) Makower is “directed to the use of flexible catheters” not “RF energy delivery devices”; (2) Makower’s Figure 5H device “is entirely hypothetical” because the parties’ otolaryngology experts had not used such a device in their years of clinical practice; and (3) “the design of Makower’s hypothetical device is fatally defective.” *Id.* at 34–46. We address these arguments in turn.

excellent recovery and efficacy) at 1 and 2 years after the surgery was significantly improved, with statistical significance.” *Id.*

Patent Owner’s argument that Makower is “directed to” the use of flexible catheters and not “RF energy devices” is legally and factually flawed. PO Resp. 34 (arguing that “almost no detail” is provided about the embodiment in Fig. 5H); *see also id.* at 50–54 (arguing “Makower is directed to improved ear, nose, and throat procedures using a *flexible* catheter” compared to previous instrumentation using more rigid shafts) (citing Ex. 2003 ¶¶ 77–85 (testifying “Makower Is Focused On Flexible Catheters”)). The law requires a POSA consider all the art’s relevant teachings and the evidence here shows that Makower is directed to and teaches more than “flexible catheters.”

Even if we accepted Patent Owner’s characterization of Makower, in an obviousness inquiry, the prior art’s teachings are not limited to what a particular reference is “directed to” or the problems the reference purports to solve. *Belden Inc. v. Berk-Tek LLC*, 805 F.3d 1064, 1076 (Fed. Cir. 2015) (faulting the Board’s disregard of certain disclosure in the art because “[a] reference must be considered for ***everything it teaches*** by way of technology and is not limited to the particular *invention* it is describing and attempting to protect”) (emphasis added); *In re Lemelson*, 397 F.2d 1006, 1009 (CCPA 1968) (“The use of patents as references is not limited to what the patentees describe as their own inventions or to the problems with which they are concerned. They are part of the literature of the art, relevant for all they contain.”). And Makower undisputedly teaches an electrosurgical device of Figure 5H and, in related disclosures, that this device can be inserted into the nasal cavity to apply current to electrodes “causing tissue adjacent to the struts . . . to be cauterized or heated.” *See, e.g.*, Ex. 1005, Fig. 5H, ¶ 187 (teaching, *inter alia*, electrodes on the struts may be “bipolar” or “monopolar”); *see also id.* ¶ 164 (teaching devices in Figs. 5A–5Y may

include components such as “radiofrequency electrodes” and that “designs and constructions of such components . . . are w[e]ll known in the art”).

Makower is not, in fact, directed only to flexible catheters. Indeed, as pointed out by Petitioner (Pet. Reply 10–11), Makower describes its “present invention” as including “a variety of [working] apparatus that may be inserted into the nasal cavity, paranasal sinus, [and] nasopharynx . . . to perform diagnostic or therapeutic procedures.” Ex. 1005 ¶¶ 161–162 (disclosing “working devices (and/or the catheters used to deliver them) may be inserted through lumens of the occluder & access devices”); *see also id.* ¶¶ 160 (heading of section on “Working Devices” for, e.g., “Ablating . . . Soft Tissue”), 164 (introducing “working devices” of Figs. 5A–5Y), claim 1 (method for “diagnosing and/or treating sinusitis or another disorder affecting the nose” that includes “advancing at least one working device” through a port device to “a location within the nose, nasopharynx or paranasal sinus” and “using the working device to perform a diagnostic or therapeutic procedure”). Moreover, there is nothing incompatible about Makower’s flexible catheters and Makower’s working devices cited by Petitioner. *See, e.g.*, Ex. 1005 ¶ 161 (“These working devices may be *delivered through or incorporated into flexible catheters* or flexible rod-like shafts” where such “flexible construction” allows the devices to be “delivered and positioned” as desired) (emphasis added); *see also id.* ¶ 187 (describing the electrosurgical device of Figure 5H as “comprising a *flexible shaft* 210 (e.g., a *catheter* or solid shaft)”) (emphasis added); Ex. 1098 ¶¶ 75–76 (testifying a POSA “would have understood that flexible catheters and radiofrequency devices such as Figure 5H of Makower can be used together”).

Patent Owner's contention that Makower's Figure 5H is a "hypothetical" device that neither clinical expert has seen nor used in practice fares no better in undercutting Petitioner's obviousness challenge. PO Resp. 34 (citing Ex. 2002 (Bu-Saba Tr.), 303:13–18 (answering "[n]ot that I recall" to whether he had seen a device "[e]xactly like Figure 5H" in practice); Ex. 2003 ¶ 89. Makower is a published U.S. patent application; we presume that its teachings are enabled. *Amgen Inc. v. Hoechst Marion Roussel, Inc.*, 314 F.3d 1313, 1354–1355 (Fed. Cir. 2003) (holding claimed and unclaimed subject matter in prior art patents is presumptively enabled); *In re Antor Media Corp.*, 689 F.3d 1282, 1288 (Fed. Cir. 2012) (extending presumption to prior art printed publications).¹⁰ Makower expressly teaches that the Figure 5H electrosurgical device is usable in the nasal cavity (and other areas such as a meatus (spaces beneath concha), ostium (opening of a sinus), and the interior of a sinus, etc.) to apply current and heat/cauterize adjacent tissues. Ex. 1005 ¶ 187; *see, e.g.*, Ex. 2003 ¶¶ 41–42 (describing internal nasal anatomy); Ex. 1064, Figs. 1–3 (same); Ex. 2001, 239–242 (same). Patent Owner cites no authority for the proposition that the art's enabled teachings are relevant in the obviousness analysis only if its

¹⁰ Patent Owner conceded at the oral hearing that Makower is presumptively enabled and it did not brief whether Makower is non-enabled. Tr. 54:5–55:9; *see also* Paper 39 (Patent Owner's Demonstrative Exhibits), 60 (describing as "*False*" the notion that Patent Owner was arguing "Inoperability" of Makower's device; instead "the reasonable expectation of success standard was employed"); PO Sur-reply 16 n.9 ("Petitioner mischaracterizes the POR as making an inoperability argument."). *See Apple Inc. v. Corephotonics, Ltd.*, 861 F. App'x. 443, 450 (Fed. Cir. 2021) ("[R]egardless of the forum, prior art patents and publications enjoy a presumption of enablement, and the patentee/applicant has the burden to prove nonenablement for such prior art.").

embodiments have been commercialized or actually used by testifying experts. *Cf. Galderma Labs., L.P. v. Tolmar, Inc.*, 737 F.3d 731, 737 (Fed. Cir. 2013) (“Nothing in the statute or our case law requires [one] to prove obviousness by starting with a prior art commercial embodiment and then providing motivation to alter that commercial embodiment”).

Patent Owner contends that Makower’s “hypothetical” device is “fatally defective.” PO Resp. 35–45. This argument arises from two premises urged by Patent Owner. The first is that the electrodes used with Makower’s device are “ring electrodes, i.e., rings that wrap around the surface of the struts of Makower’s device” and, thus, are “configured to emit energy about their entire circumference.” *Id.* at 36–39 (citing Ex. 2003 ¶¶ 91–97; Ex. 2005 ¶¶ 64–71, 76). Second, Patent Owner argues that a POSA “would recognize that the nasopharyngeal passage would include bodily fluids, such as mucus or blood, and/or saline (e.g., if the passage was irrigated prior to treatment)” and these “highly-conductive” fluids in the “hollow interior” (i.e., space between struts of Makower’s device) would cause ions to flow into the fluid “and not the target tissue.” *Id.* at 38–39 (annotating Makower’s Fig. 5H to add fluid (colored yellow) filling the cavity/passage into which the device is advanced). For these reasons, Patent Owner contends, a POSA “would not have reasonably expected [Makower’s device] to ablatively treat target tissue.” *Id.* at 40; PO Sur-reply 11–15.

Patent Owner’s argument is unavailing. *First*, Patent Owner’s interpretation of Makower is directly at odds with Makower’s express (and presumptively-enabled) teachings. Makower teaches, specific to Figure 5H, that, after the device is advanced into the nasal cavity, “current is applied to the electrodes 216 **causing tissue adjacent to the struts** 214 to be cauterized or heated.” Ex. 1005 ¶ 187 (emphasis added). Patent Owner’s argument

requires we agree that Makower's device would *not work* in the manner Makower teaches the device *does work*. In its Sur-reply, Patent Owner emphasizes that Makower itself recognizes that fluids may be present in the nose. PO Sur-reply 12–13 (citing Ex. 1005 ¶¶ 132, 135). Even so, it is telling that Makower does not identify any material impediment that such fluids would have to delivering energy to target tissues with its devices.¹¹ If anything, the fluid disclosures of Makower cited by Patent Owner suggest that skilled artisans knew how to manage fluid (and remove it, if necessary) at the site of treatment. *See, e.g.*, Ex. 1005 ¶ 135 (teaching that infusion and aspiration lumens can be used for “concurrent infusion of an irrigation fluid and suctioning of the irrigation or other fluid from the operative field” and describing the same as “useful for clearing blood or debris from the operative field” and to facilitate unobstructed viewing of the anatomical structures and placement of the working devices of Figs. 5A–5Y).

Second, we credit the testimony of the experts that, on balance, further confirms that skilled clinicians knew how to manage and remove fluids to the extent necessary. Dr. Bu-Saba testifies that typical steps for RF ablation procedures (even before 2006) included positioning the patient in a seated or slightly reclined position to facilitate the flow of any mucus or fluids away from the nasal cavity, application of topical decongestants, and “[i]n some circumstances,” if mucus or blood was still in the treatment area, “a

¹¹ Nor does Patent Owner identify any disclosure in any other technical literature (either before or after 2006) expressing concerns that electrolytic fluids at the operative site could not be managed and/or would prevent targeted application of RF ablation energy to tissues at that site. *See, e.g.*, Ex. 2003 ¶¶ 91–97 (testifying about the hypothetical “fluid” problem, but identifying no independent recognition of such a problem in any corroborating reference); Ex. 2005 ¶¶ 66–76 (same).

physician would suction the mucus out of the treatment area to optimize visualization of the surgical site.” Ex. 1098 ¶¶ 81–89; Pet. Reply 14–15. Patent Owner’s expert, Dr. Levine, concurs that skilled clinicians knew how to remove mucus and other fluids, testifying that “[i]f [mucus] was obstructing my view” and “[i]f it was this thick . . . I might have to mechanically remove it with forceps, I might have to suction it, I might have to irrigate the nose with a saline solution to wash out the nose.” Ex. 1089 (Levine Tr.), 83:3–84:6 (admitting residual saline could either be swallowed or suctioned out); *see also id.* at 86:1–16 (testifying a patient may be positioned such that by “[v]irtue of gravity . . . fluid would fall to the bottom”).

Third, Patent Owner presumes that Makower’s Figure 5H device is limited to a use of “ring” electrodes. PO Resp. 35–37 (arguing electrodes “appear” to be ring electrodes from the drawing). But Makower’s disclosure is not so narrow. The accompanying disclosure for Makower’s Figure 5H embodiment is broader—indicating that “electrodes 216” may be “bipolar, monopolar” or use “other suitable” energy. Ex. 1005 ¶ 187. Makower also directs the skilled artisan to suitable and “w[e]ll-known” components for its 5A–5Y working devices, including “radiofrequency electrodes,” described in additional background references. *Id.* ¶ 164 (citing, e.g., “U.S. Pat. No. 5,255,679 (Imran)” (Ex. 1068)). Imran, for example, describes a basket assembly ablation device that uses a plurality of surface electrodes located along expandable arms or struts. Ex. 1068, Fig. 1, 2:57–3:7 (describing and depicting electrodes 23); Pet. Reply 12–13. Altogether, we find that Makower’s teachings should be interpreted broadly, not narrowly, and not limited to “ring” electrodes as urged by Patent Owner.

Patent Owner contends Petitioner’s reliance on Imran and Imran’s Figure 1 embodiment in the Reply is untimely and improper. PO Sur-reply 10. We disagree. Petitioner’s reliance on Imran is fairly responsive to Patent Owner’s position that a POSA would interpret Makower’s teachings as requiring use of “ring” electrodes, and moreover, we find that there is no unfairness to Patent Owner because Petitioner expressly cited Imran’s Figure 1 and related disclosure *in the Petition* when addressing dependent claim 9. Pet. 86 (citing Ex. 1068, Fig. 1, 3:2–7; Ex. 1002 ¶¶ 204–206 (reproducing Imran’s Figure 1 as a non-limiting example of known electrode components “disposed on the lateral side of a strut” of an ablation treatment device and testifying “Makower contemplates . . . implementation of laterally disposed electrodes” like shown in Imran)).

Patent Owner also points out that Imran’s ablation catheter is designed for chambers of the heart. PO Sur-reply 10. But this only underscores that a skilled artisan would recognize device components (including RF electrodes for ablation) have broader therapeutic uses—insofar as Makower, a reference ostensibly about treating sinusitis and other diseases of the ears, nose, and throat, directs the reader to components of Imran’s devices for ablating tissues in the heart. *Compare* Ex. 1005, Abstr., *with* Ex. 1068, Abstr; Ex. 1005 ¶ 164. Again, we find that a POSA would not interpret Makower’s teachings about the 5H device as requiring ring electrodes.

Lastly, even if Makower required ring electrodes, we are unpersuaded that its device would be “fatally defective.” As Petitioner notes, the ’675 patent’s Figure 5B “uses a ring electrode, is covered by the claims, and is operable.” Pet. Reply 13 (citing Ex. 1089 (Levine Tr.) 195:4–196:6 (agreeing “Figure 5B shows ring electrodes,” is “operable . . . [a]ccording to this description” in the patent, and that the patent conveys no “challenges

with applying energy to nasal tissue using the device in Figure 5B”); Ex. 1001, Fig. 5B (showing ring electrodes with an energy field distributed around the distal body of the device); Ex. 1098 ¶¶ 89–91. If, as urged by Patent Owner, mucus and other fluids preferentially draw ions away from delivery into adjacent tissues and into the fluid, we do not see how this same phenomenon would not make the patent’s Figure 5B embodiment “fatally defective,” contrary to Dr. Levine’s testimony or any indication of such a problem in the patent (much less how it is solved). Patent Owner argues the challenged patent is not prior art and cannot be used to make Makower’s device functional. PO Sur-reply 15. That is not the manner for which the patent is cited here. Makower teaches that its device *is functional* and *delivers energy into adjacent tissues* (and is presumptively enabled for the same as already explained). The ’675 patent is cited because it highlights an inconsistency in Patent Owner’s position—and suggests Patent Owner’s argument about an absence of functionality or operability with Makower’s device lacks merit. Tr. 32:25–33:21 (“Patent Owner has not explained how Figure 5B could be operable and covered by the claims, which they say it is, and Makower would not.”).¹²

Beyond the arguments addressed above, Patent Owner also questions whether Makower’s Figure 5H device with its allegedly “rigid” struts would be capable of allowing contact between the electrodes and nasal cavity

¹² The lack of a “hollow interior” in the Figure 5B embodiment does not change our analysis. PO Sur-Reply 15. Either fluid surrounds the electrodes on the cavity side of the device and allegedly draws energy away from the tissue side of the device—or not, and energy is still applied to the tissues. If the supposed “fluid” issue arises at all, it would seemingly arise whether it is Makower’s strut(s) or Figure 5B’s catheter that contacts tissue.

tissues desired for targeting. PO Resp. 44–45 (arguing that, in the depiction of Fig. 5H, at least one of the four electrodes is not contacting tissue). This argument is weak and notably unsupported by any cited testimony from Patent Owner’s declarants. The only persuasive testimony on this issue comes from Dr. Bu-Saba, who consistently testifies that a POSA would have reasonably expected that Makower’s device could be used on the nasal cavity tissues targeted by Fang. Ex. 1002 ¶¶ 185, 188–190; Ex. 1098 ¶¶ 99–101 (explaining that Makower teaches its device may be used to treat tissues throughout the nasal cavity and nasopharynx and that the Fig. 5H depiction was not specific or limiting as to anatomical location). Patent Owner also cites no support for its suggestion that Makower’s struts are necessarily “rigid”—rather, the evidence suggests otherwise because Makower teaches the struts “may be collapsible and expandable.” Ex. 1005 ¶ 187. We also reject the notion that a skilled artisan could not manipulate and orient the device as desired such that the electrodes contact target tissues. Ex. 1005 ¶ 161 (teaching the devices may comprise flexible and/or rigid portions as needed “to facilitate their control and guidance to the appropriate region”).

Patent Owner also criticizes Petitioner’s reliance on Fang. PO Resp. 42–45. According to Patent Owner, the Board should reject Petitioner’s position about the alleged reasonable expectation of success because “Fang’s SHP-I device is an unknown quantity,” Dr. Bu-Saba had not previously used Fang’s device, and allegedly he could not decipher Fang’s results. *Id.* at 42–43 (citing, e.g., Ex. 2002, 182:4–10).¹³

¹³ We discuss Patent Owner’s additional criticisms of Fang in the section below related to the asserted lack of reasons for combining Makower and Fang, consistent with Patent Owner’s ordering of its arguments. *See, e.g.*, PO Resp. 46–49 (arguing, *inter alia*, that Fang’s results are inconclusive).

We are unpersuaded. As discussed above and as explained by Petitioner and Dr. Bu-Saba, Makower teaches that its catheter device is capable of providing RF ablation energy to heat/cauterize tissues within the nasal cavity, and Fang, which also applies RF ablation energy, provides further disclosure about the nasal cavity tissues to target for ablation and treatment of rhinitis and the anticipated effects of doing so. Fang discloses sufficient device and treatment details relevant for a reasonable expectation of success under Petitioner’s theory. Pet. Reply 15; Ex. 1098 ¶¶ 96–98 (testifying, *inter alia*, “Fang’s teachings are consistent with what was known in the art, and as a result, a [POSA] would have reasonably expected that Makower’s device would achieve Fang’s results when applying ablation energy to tissue in the nasal cavity or nasopharynx.”); Pet. 71–77; Ex. 1002 ¶¶ 46, 179–181, 188–190; Ex. 1006, 876–877.¹⁴ Furthermore, claim 1 requires no specific efficacy measure or endpoint. Ex. 1098 ¶ 111 (testifying the “claims at issue do not require any efficacy measure or particular degree of efficacy, and the ’675 patent does not describe any such efficacy measures or requirements for treating rhinitis”); Ex. 1089 (Levine Tr.), 184:21–186:7 (admitting the challenged patents do not define any efficacy requirements for the treatment of rhinitis, or for the reduction of congestion or mucus production); Tr. 46:12–19 (Patent Owner agreeing the

¹⁴ *Donner Tech., LLC v. Pro Stage Gear, LLC*, 979 F.3d 1353, 1361 (Fed. Cir. 2020) (“A [POSA] might reasonably choose to consult a reference even if she would not understand every last detail of that reference, so long as she would understand the portions of the reference relevant to solving her problem well enough to glean useful information”).

claims require no particular efficacy degree or measure).¹⁵ Given the level of generality in the claims, more was not needed from Petitioner to establish a reasonable expectation of success in arriving at that claimed subject matter.

For all the reasons above, we determine that the evidence supports Petitioner’s contention that a POSA would have reasonably expected success in combining the prior art to arrive at the subject matter of claim 1.

(2) *Alleged Absence of a Motivation to Combine Makower and Fang*

Patent Owner argues that Petitioner failed to establish a motivation to combine Makower and Fang. PO Resp. 45. According to Patent Owner, a POSA would not have used Makower’s RF-ablation device according to Fang’s method of treating rhinitis with RF ablation because “Fang’s results were inconclusive.” *Id.* at 45–48. Patent Owner argues that Makower is “directed to improved ear, nose, and throat procedures using a *flexible* catheter,” not RF-treatment devices. *Id.* at 48–53. And, Patent Owner argues, Petitioner’s reasons for combining Makower and Fang are biased by hindsight. *Id.* at 53–65; PO Sur-reply 17–21. We address Patent Owner’s arguments below.

(a) *Fang’s “Inconclusive Study”*

At the outset, it is uncontested that Makower and Fang are analogous art to the ’675 patent. *See* Pet. 62; PO Resp. 62; PO Sur-Reply 17.

¹⁵ Counsel’s statement purporting to qualify Patent Owner’s agreement “for purposes of written description” is not a valid qualifier. Tr. 46:12–19. The scope of the claims does not change depending on whether the issue is written description/priority or obviousness.

Patent Owner argues that Fang’s study lacks clarity and Patent Owner characterizes Fang’s results as “inconclusive.” PO Resp. 45–48. According to Patent Owner, among the 63 patients in Fang’s experimental group, some unknown number of those patients received additional surgical treatments (beyond septum resection and RF ablation), thus undermining Fang’s “hypothesis” that RF ablation of the targeted tissues provided a beneficial therapeutic effect for rhinitis. *Id.* at 46–47 (citing Ex. 1006, 876 (disclosing that “uncinate process resection, sinus opening, . . . [or] nasal polypectomy were performed for those who had hypertrophy of the uncinate process, sinusitis, . . . or nasal polyps”)). Patent Owner argues that “[a]ny of these additional procedures may have been wholly or partially responsible for the end results Fang’s patients reported.” *Id.* at 47 (citing Ex. 2003 ¶¶ 72–74). Patent Owner also argues that Fang itself does not expressly disclose the diagnostic or efficacy criteria for its study, and instead, Patent Owner points out that Fang directs the reader to another document (“citation [1]”) for that criteria, which document Petitioner did not submit with its Petition.¹⁶ *Id.* Patent Owner suggests the purported lack of clarity and problems with

¹⁶ This document (Ex. 1113 (Haikou, *Attachment: Diagnostic Criteria and Efficacy Evaluation Criteria of Allergic Rhinitis*, 33:3 CHIN. J. OTORHINOLARYNGOL. 134–135 (June 1998) (“Haikou”)), which Petitioner filed with its Reply, is the subject of Patent Owner’s pending Motion to Exclude. *See* Ex. 1114 (uncontested translator’s declaration). We acknowledge Patent Owner’s argument regarding the alleged absence of detailed diagnostic or efficacy criteria in Fang itself, but the claims require no specific efficacy measure or endpoint. And Petitioner does not and need not rely on Fang or Haikou for disclosure of criteria that are not required by the claims. We do not rely on Haikou to reach our determinations regarding the patentability of the challenged claims.

Fang’s study design may explain why “Fang has been cited a single time” despite its publication over 19 years ago. *Id.* at 48 (citing Ex. 2008).

Having considered Patent Owner’s argument, we agree with Petitioner that a POSA would not have dismissed Fang’s reported positive results. Pet. Reply 17–20. *First*, the law does not require “conclusive proof” of efficacy to demonstrate obviousness. *Acorda Therapeutics, Inc. v. Roxane Labs., Inc.*, 903 F.3d 1310, 1333 (Fed. Cir. 2018) (“This court has long rejected a requirement of conclusive proof of efficacy for obviousness”) (internal quotation marks and alteration in original); *but see* Ex. 2003 ¶ 71 (Dr. Levine opining a POSA “would need to see *conclusive proof* that RF energy alone was responsible for the improvement in perennial allergic rhinitis”) (emphasis added). That is especially so where the claims here, insofar as they recite efficacy at all, only do so at the most generic level as we discussed above. *See, e.g.*, Ex. 1098 ¶ 111 (testifying the claims require no particular degree or measure of efficacy).

Second, we agree with Petitioner that Fang teaches RF ablation of the parasympathetic nerves in the nasal cavity/nasopharynx improved rhinitis outcomes. Pet. Reply 18. Indeed, Fang reported statistically significant improvements between the control group (no ablation) and the study group (with ablation). Ex. 1006, 876 (disclosing that “[c]ompared with patients in the control group . . . , the total efficacy rate [among the experimental group] . . . was significantly improved, with statistical significance.”); *see also id.* (concluding nasal endoscope “with radiofrequency ablation may achieve good therapeutic effects on perennial allergic rhinitis”). Even considering that some patients may have received additional surgical interventions for additional conditions (e.g., nasal polyps), we are unpersuaded that a skilled artisan would have dismissed Fang’s favorable results. Ex. 1098 ¶¶ 104–

106, 116–120 (testifying, *inter alia*, it is common for clinicians to treat multiple pathologies simultaneously and that a POSA would have considered (and not discounted) Fang’s attribution of positive rhinitis treatment results to the RF ablation technique Fang applied).

Third, although Patent Owner criticizes parts of Fang’s experimental method and the reference to diagnostic/efficacy criteria in a document outside Fang’s four-corners, Patent Owner “does ***not dispute*** Fang was right about” the benefit of RF ablation in treating rhinitis. Pet. Reply 18–19. We find that a POSA would, on balance, have known or at least reasonably expected that Fang was correct—as “Fang’s findings are consistent with multiple undisputed prior-art references.” *Id.*; Ex. 1002 ¶¶ 46 (discussing background references), 190 (discussing Chen); Pet. 14–15 (asserting “[a] distinct but well-known technique was to use **ablation** devices to deliver energy to destroy cells that cause rhinitis symptoms” (citing Fang, Chen, and Arora (Ex. 1054, 80)), 79, 83 (discussing Chen in the context of motivation and reasonable expectation of success). Chen, for instance, discloses that “radiofrequency thermocoagulation of the sphenopalatine nerve” reduced the symptoms of rhinitis, consistent with Fang’s results. Ex. 1044, 239–240 (describing the technique as “highly effective, easy to perform . . . [and] a good way to treat PAR [(perennial allergic rhinitis)]”); *see also* Ex. 1054, 80 (“[c]ryodestruction of the nasal branches of the Vidian nerve . . . in cases of vasomotor rhinitis”); Ex. 1045, 341–342 (“[RF] treatment . . . can cause varying degrees of damage to the deep ethmoidal nerve . . . thus achieving the goal of treating allergic rhinitis”); Ex. 1098 ¶ 98 (testifying it was already known “that ablating parasympathetic nerves, even with various forms of ablation such as [RF] ablation or cryotherapy, treats rhinitis” (citing references)). Considered in the context of Chen and other background art

cited by Petitioner, which Patent Owner does not address in its papers, the criticisms about lack of detail in Fang fall flat. Tr. 51:20–53:8.¹⁷

Lastly, Patent Owner’s contention that Fang has been cited only once in the technical literature is incorrect and unpersuasive. In response to Patent Owner’s argument, Petitioner provides evidence indicating that Fang has been cited at least seven times, including in the Hytönen reference that Patent Owner alluded to, but did not submit. Pet. Reply 19–20 (citing Ex. 1103). Hytönen cites positively to Fang’s results and urges that clinical trials on nasal RF ablation treatment should proceed. Ex. 1112, 1260–1261, 1264 (noting the statistically significant “SS” results reported in Fang

¹⁷ While we need not resolve Patent Owner’s priority arguments (for Grounds 1–3) raised in this proceeding, we note some of those arguments are in tension with its criticism about Fang. When arguing the alleged priority disclosures for the ’675 patent provide sufficient written-description support for treating rhinitis using RF-energy only (i.e., without any toxin delivery), Patent Owner repeatedly argued that a POSA would have simply understood that ablating target tissues results in rhinitis treatment—despite the patent disclosures having no working example or experimental data to support this position. *See, e.g.*, Paper 39, slide 17 (“Undisputed: Ablating Goblet Cells Treats The Claimed Conditions”); Tr. 43:7–44:7 (“[I]t’s understood by persons skilled in this art as evidenced by their own expert’s testimony and our expert’s testimony that when you apply irreversibly porated energy to goblet cells, you are going to treat these conditions”); Ex. 1089, 3:7 (Dr. Levine agreeing the patents provide *no data* about treating rhinitis with energy in the absence of toxin). When arguing nonobviousness, however, Patent Owner seemingly demands detailed efficacy data from Fang on the use of RF energy for rhinitis treatment that is found nowhere in the ’675 patent itself. *In re Epstein*, 32 F.3d 1559, 1568 (Fed. Cir. 1994) (holding “the Board’s observation that appellant did not provide the type of detail in his specification that he now argues is necessary in prior art references supports the Board’s finding that one skilled in the art would have known how to implement the features of the references”).

(among other studies) and recommending initiation of double-blind, placebo-controlled randomized trials).

In sum, Patent Owner’s critiques of Fang would not have discouraged the skilled artisan from considering its teachings, nor do they detract from the reasons given by Petitioner for combining Makower and Fang.

(b) *Whether Makower is “Directed to”
Flexible Catheters*

Patent Owner argues Makower is “directed to flexible catheters” and that Petitioner provides “no reason” why a POSA would have looked to Makower for energy delivery devices. PO Resp. 48–53 (citing, e.g., Ex. 1005 ¶¶ 36–37, 40–42, 132–143; Ex. 2003 ¶¶ 77–85).

We addressed substantially the same argument above, and explained why it was unpersuasive. *See supra* Section III.F.1.b & b.1. Makower is undisputedly analogous prior art, and we presume that a POSA is aware of all such relevant prior art. *In re Rouffet*, 149 F.3d 1350, 1357 (Fed. Cir. 1998) (explaining that the law “presumes that all prior art references in the field of the invention are available to this hypothetical skilled artisan”); *In re Lemelson*, 397 F.2d at 1009 (holding all relevant disclosure must be considered and a reference’s teachings are not limited to the “problem” solved by the reference). Makower is not limited to “flexible catheters” and, indeed, it describes and claims working devices—such devices include an RF-ablation catheter device designed for insertion into the nasal cavity such as described and shown in Makower’s Figure 5H. *See, e.g.*, Ex. 1005 ¶¶ 160–162, 164, 187, claim 1, Fig. 5H. Makower also teaches flexible catheters can be used with or incorporated into such working devices to facilitate safe and accurate placement at the appropriate region for which treatment is desired. *See, e.g., id.* ¶¶ 161, 187.

We determine the evidence, including the teachings of the references themselves, favors Petitioner’s argument that a POSA would have been motivated to use Makower’s RF ablation device to target the regions of the nasal cavity identified by Fang and, thus, treat rhinitis. On that issue we credit Dr. Bu-Saba’s testimony, which is supported by the evidence of record, over the competing testimony of Dr. Levine. *See, e.g.*, Ex. 1002 ¶¶ 169–179; Ex. 1098 ¶¶ 124–130 (addressing reasons why a skilled artisan would have considered Makower’s RF ablation device and explaining how the opinion is grounded in Makower’s and Fang’s actual teachings and the artisan’s background knowledge); *see, e.g.*, Ex. 2003 ¶¶ 63–88 (testifying a POSA would not have combined Makower and Fang due to Makower’s alleged focus on flexible catheters and Fang’s supposedly flawed study—reasoning that we reject as unpersuasive).

(c) *Alleged Hindsight and Lack of
Reasons to Combine the Art*

Patent Owner contends Petitioner’s reasons for combining Makower and Fang are tainted with hindsight. PO Resp. 53–63. Patent Owner invokes a variety of contentions that allegedly reveal this flaw in Petitioner’s challenge. Patent Owner contends a skilled artisan would not have looked to Makower’s device for damaging nerves because Makower discloses that its device shrinks tissues as a means of reducing obstructions in the nasopharyngeal passage. *Id.* at 54 (citing Ex. 1005 ¶ 92; Ex. 2003 ¶ 45–49). Patent Owner argues Makower and Fang have different objectives; Makower is allegedly focused on flexible devices as a way to avoid bloody or painful treatments that historically arose from use of rigid devices, and Fang’s methods included “destructive surgical methods,” including septum surgery. *Id.* at 55–56 (citing Ex. 1006, 877; Ex. 2003 ¶¶ 89–98). Patent Owner

contends the electrode arrangement of Makower’s device could cause undesired ablation to tissues beyond those targeted for treatment. *Id.* at 56–58 (asserting Makower’s Fig. 5H shows a device positioned to cauterize tissue on both sides of a passageway and, if such a device with oppositely disposed struts were placed against the lateral nasal wall (i.e., tissues targeted by Fang), the septum opposite the lateral wall might be cauterized too). And, Patent Owner contends, if anything, a skilled artisan would select other devices of Makower—like the Figure 5A device, which includes only a single side opening that could be positioned against adjacent tissues—rather than the two-sided treatment device shown in Figure 5H, which comes with alleged risks of ablating and potentially perforating the septum. PO Resp. 59–60 (citing Ex. 1005 ¶ 166, Fig. 5A).¹⁸

Patent Owner’s contentions are unavailing. We find that a skilled artisan would have considered Makower for nasal disorder treatments. *See, e.g.*, Ex. 1005 ¶ 1 (describing field of invention as related to catheter based devices and methods for treating sinusitis and other nasal disorders). Insofar as Makower’s Figure 5H device can be used for tissue shrinking, it shrinks tissues through the application of ablation energy. Ex. 1005 ¶¶ 92, 164, 187; Ex. 1098 ¶ 132 (testifying that tissue-shrinking electrodes include multiple bipolar electrodes delivering ablation energy to tissues). Moreover, treatments aimed at reducing obstructions and treatments aimed at reducing

¹⁸ That the Examiner cited Makower during the prosecution history of the ’675 patent does not factor persuasively one way or the other into our analysis on this trial record. *See* Pet. 17; Pet. Reply 20–21 (noting that the Examiner identified Makower as the “closest prior art” during prosecution); Ex. 1022, 122; PO Resp. 62 (noting the Examiner ultimately allowed the claims, finding Makower did not teach disrupting neural activity to reduce mucus secretions in the nasal cavity).

mucus secretions are not incompatible or mutually exclusive. Ex. 1098 ¶¶ 133–134 (testifying blockages and mucus production are both symptoms of rhinitis, and treatments also involved reduction of nasal blockages); Ex. 1107 (Levine article), 453 (describing rhinitis treatment using laser coagulation to reduce turbinate and decrease mucus-secreting tissues).

We do not agree that Makower and Fang have diametrically opposed objectives, as argued by Patent Owner. PO Resp. 55–56. Both references teach or suggest treating nasal conditions by application of RF-ablation energy. Ex. 1006, 876–877; Ex. 1005 ¶¶ 1, 43, 132, 164, 187. Makower does not teach that cutting or similar surgical interventions are inconsistent or inappropriate with its devices and methods. *See* Ex. 1005 ¶ 43 (disclosing that devices of the invention may be used for “remodeling,” “cutting,” and “debulking” nasal tissues “such as nasal polyps” and other “enlarged” or “abnormal” tissues). Drs. Levine and Bu-Saba agree that “it’s not unusual to combine several procedures in practice” when treating nasal conditions. Ex. 1089, 89:12–91:8 (testifying, e.g., that a clinician may correct a deviated septum during the same procedure in which a turbinate is reduced in size); Ex. 1098 ¶¶ 136–138 (testifying “combining multiple procedures was (and remains) common”). And nothing in challenged claim 1 forecloses surgical or other therapeutic interventions above and beyond RF ablation.

We also disagree with Patent Owner’s contention that a POSA would not have used Makower’s Figure 5H device based on concerns about ablation of undesired tissues. As Petitioner argues, a POSA would simply adjust the device’s positioning or its RF parameters, such as “by selectively activating electrodes contacting target nasal-wall tissue.” Pet. Reply 22–23 (citing Ex. 1005 ¶¶ 43, 161, 164, 187); Ex. 1098 ¶¶ 139–143; *KSR*, 550 U.S. at 421 (explaining that a skilled artisan is “not an automaton”). That a

POSA might have favored and, therefore, selected a different device described by Makower (i.e., the embodiment of Figure 5A) for the surgical procedures in question is not decisive. A petitioner need not show a motivation to pursue only the best or most obvious solution. *Intel Corp. v. Qualcomm Inc.*, 21 F.4th 784, 800 (Fed. Cir. 2021).

For the reasons above, we find that the preponderance of the evidence supports Petitioner’s contention that a POSA would have combined Makower and Fang in the manner proposed.

c) Conclusion on Claim 1

Upon considering the full trial record, we find by a preponderance of the evidence that the combination of Makower and Fang teaches or suggests the method of claim 1, and that the skilled artisan would have had reason to combine the references as proposed by Petitioner and done so with a reasonable expectation of success. We, therefore, conclude that claim 1 would have been obvious.

2. Claim 17

Similar to claim 1, independent claim 17 is directed to “a method of treating a nasal condition of a patient” in three steps: 1) “positioning a distal portion of a treatment device proximate to target tissue in a nasopharyngeal passageway of a patient;” 2) “administering therapy to the target tissue with the treatment device” to “alter” the target tissue “such that congestion in the nasopharyngeal passageway is reduced;” and 3) “administering therapy” comprising “delivering energy to the target tissue.” Ex. 1001, 2822–31. Claim 17 differs from claim 1 primarily in the result achieved by treatment. Claim 1 requires “administering therapy to the target tissue with the treatment device so as to disrupt neural activity *such that mucus secretion is*

reduced in the nasopharyngeal passage,” while claim 17 requires “administering therapy to the target tissue with the treatment device to alter nasopharyngeal cells such that congestion in the nasopharyngeal passage is reduced.”

There is no dispute between the parties, however, specific to claim 17. Petitioner’s challenge of claim 17 relies heavily on its analysis of claim 1, and Patent Owner does not argue the patentability of claim 17 separately. *See* Pet. 82–83; *see also* PO Resp. 32–45. Thus, for the same reasons discussed above with regard to claim 1, we find by a preponderance of the evidence that the combination of Makower and Fang teaches or suggests the method of claim 17, and that the skilled artisan would have had reason to combine the references as proposed by Petitioner and done so with a reasonable expectation of success. We, therefore, conclude that claim 17 would have been obvious.

3. Claim 22

Claim 22 depends indirectly from claim 17 and adds that “the energy delivery element is configured to concentrate energy laterally relative to the distal portion of the treatment device.” Ex. 1001, 28:41–43. The parties’ core dispute is whether Makower teaches or suggests a device that uses bipolar electrodes configured to deliver energy laterally. For the reasons that follow, we determine that Petitioner has the better position.

a) *Petitioner’s Contentions*

Petitioner contends that the combination of Makower and Fang teaches an energy transmission element (electrodes) configured to deliver energy laterally relative to the distal portion of the body as require by claim 22. Pet. 85–87; Pet. Reply 23–26. To support that contention, Petitioner

directs our attention to Figure 5H of Makower, which provides a partial perspective view of Makower's tissue shrinking electrode device. Ex. 1005 ¶¶ 92, 187; *see supra*, Section III.E.1. Petitioner contends that Makower's electrosurgical device includes bipolar or monopolar electrodes 216 that apply energy to adjacent tissue in a manner similar to the electrodes in ['675] patent Figure 5A. Pet. 86–87 (citing Ex. 1001, 16:30–58, 5A–5B; Ex. 1005 ¶ 187, Fig. 5H; Ex. 1068,¹⁹ 3:2–7; Ex. 1002 ¶¶ 200–202).

Petitioner contends that

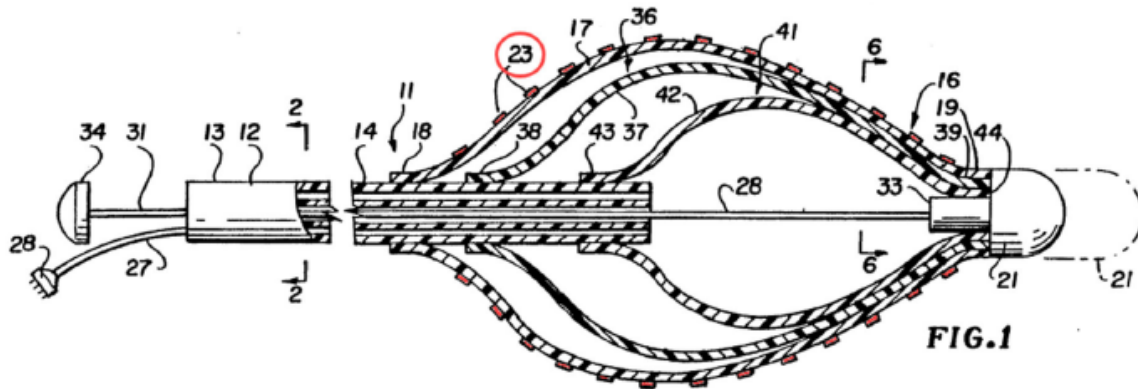
In Makower, a POSITA would have understood current travels from one bipolar electrode to another electrode to create an electrical field that travels through contacted tissue. Current (energy) in this field is delivered laterally relative to the distal portion of the device (i.e., into target tissue and away from the shaft).

Pet. 86 (emphasis omitted). Petitioner further relies on the testimony of Dr. Bu-Saba that

A person of ordinary skill in the art would have understood the concentration of current, i.e., energy, in Makower is directed laterally relative to the strut members of the electrosurgical device. In other words, a person of ordinary skill in the art would have understood that current traveling between Makower's bipolar electrodes would create electrical fields that concentrate current lateral relative to the axis of the distal portion of the RF ablation probe and into the targeted tissue.

Ex. 1002 ¶ 206. Dr. Bu-Saba also states that Makower references electrode assemblies known in the art that are capable of being incorporated into Makower's design. *Id.* In particular, Dr. Bu-Saba relies on an annotated Figure 1 of Imran, reproduced below, to support his testimony. *Id.*

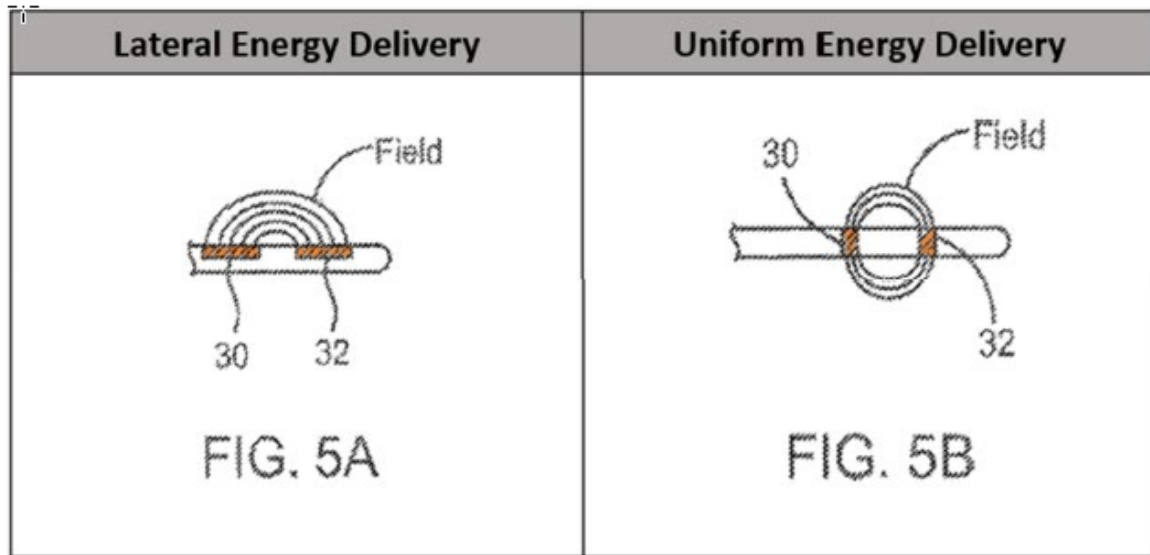
¹⁹ Ex. 1068, Imran, US 5,255,679, issued Oct. 26, 1993 ("Imran").



Imran's Figure 1 depicts an endocardial catheter for ablation having a "plurality of longitudinally spaced apart electrodes 23." Ex. 1068, 3:2–3. With reference to Imran's Figure 1, Dr. Bu-Saba states that Imran's electrodes are configured to deliver energy laterally because they "are disposed on the lateral side of a strut on the distal portion of the treatment device." Ex. 1002 ¶ 206. Dr. Bu-Saba concludes that "a person of ordinary skill in the art would have understood [from Imran's device] that current traveling from or between such electrodes would create electrical fields that concentrate current lateral relative to the axis of the distal portion of the RF ablation probe." *Id.*

b) Patent Owner's Contentions

Patent Owner contends that lateral energy delivery requires a specific electrode configuration (e.g., '675 patent, FIG. 5A). PO Resp. 65–66. To begin, Patent Owner directs our attention to annotated Figures 5A and 5B of the '675 patent, reproduced below.



According to Patent Owner, Figure 5A of the '675 patent depicts an arrangement of electrodes 30, 32 that are insulated, resulting in an electrical field concentrated in one direction—"laterally through the electrode's top surface." *Id.* at 65 (citing Ex. 2003 ¶ 109; Ex. 2005 ¶¶ 74–75). Figure 5B depicts an arrangement of electrodes that "create a more uniform electric field about the shaft of the catheter body." *Id.* at 14; Ex. 1001, 16:54–58 ("FIG. 5A depicts an arrangement of electrodes 30 and 32 that produces an electric field concentrated in a lateral direction from the catheter body whereas, FIG. 5B shows a device constructed to create a more uniform electric field about the shaft of the catheter body.").

Next, Patent Owner contends that Makower's electrode arrangement does not deliver energy laterally. PO Resp. 66. Rather, Patent Owner contends that a

Makower's ring electrodes are merely located on the surface of Makower's struts and therefore fail to include any insulating surface that would cause the electrodes direct concentrate energy delivery in a lateral direction. (Ex-2003-Levine at ¶104; Ex-2005-Bikson at ¶¶23, 76-77, 83.)

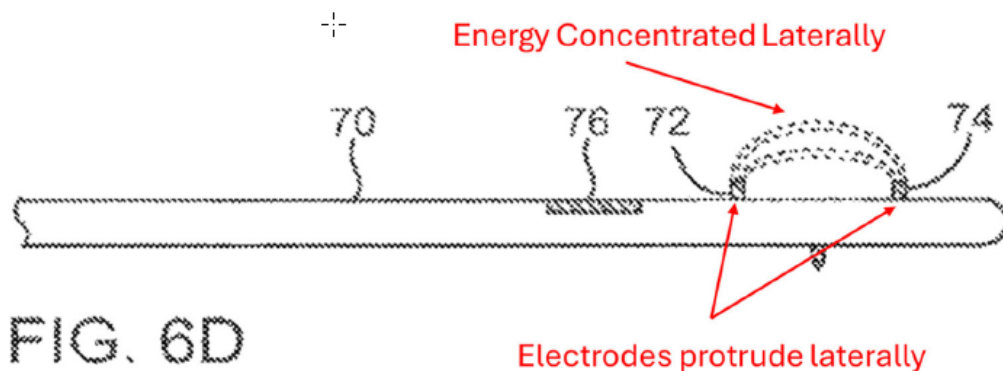
PO Resp. 66.

c) Petitioner's Reply

Petitioner responds that “Makower Figure 5H concentrates energy laterally relative to the distal portion of the treatment device (i.e., in tissue) when current travels between bipolar electrodes, . . . just like the direction energy is concentrated in '675 patent Figure 5A.” Pet. Reply 23 (citing Ex. 1098 ¶¶ 145–146; Pet. 89–90). Petitioner further contends that “a POSITA would have expected ring electrodes in Makower [to] concentrate energy laterally because air would insulate any portion not contacting tissue.” *Id.* at 24 (citing Ex. 1091, 218:9–11, 221:18–222:7; Ex. 1001, 18:11–14; 23:65–24:3; Ex. 1098 ¶¶ 147–148, 151–152).

Additionally, Petitioner contends that Patent Owner's arguments that Imran's protruding electrodes would not deliver energy laterally (Ex.2003-Levine, ¶109; Ex.2005-Bikson, ¶82; Ex.1089-Levine-Tr., 187:7-188:5), are directly contradicted by the '675 patent, which in Figure 6D depicts electrodes that “extend laterally” **and** create a lateral “energy field” akin to Figure 5A, and in claim 10 claims energy transmission elements that **both** extend **and** deliver energy laterally.

Pet. Reply 26 (citing Ex. 1001, Fig. 6D, 17:27–35, claims 1, 9–10; Ex. 1089, 190:6–9; Ex. 1098 ¶ 153). To support that contention, Petitioner directs our attention to an annotated Figure 6 of the '675 patent, reproduced below.



As described in the '675 patent, Figure 6D “depicts a catheter device 70 having electrode elements 72 and 74 that are adapted to extend laterally from the main catheter body.” Ex. 1001, 17:27–29. Petitioner’s annotations point to the field that is created between electrode elements 72 and 74 that suggest that energy is concentrated laterally despite sitting on the surface of the main catheter body. Petitioner contends that, “[i]f Figure 5H [of Makower] were implemented with Imran’s surface bipolar electrodes, it would do the same.” Pet. Reply 24.

d) Patent Owner’s Sur-reply

Patent Owner contends that

Even if Imran is improperly considered, its electrode configuration does not deliver energy laterally because those electrodes sit on a surface, i.e., are uninsulated. Not only is energy emitted in all directions from such uninsulated electrodes, but the electrolytic fluid in the nasal cavity precludes energy from being delivered laterally—it draws current away from the target tissue. (Ex-2003, ¶109; Ex-2005, ¶¶76, 81-82.)

PO Sur-reply 22. Patent Owner supports its contention with declaration of Dr. Bikson, who testifies that

Unlike the FIG. 5A electrodes in the '675 patent, Makower’s FIG. 5H electrodes are ring electrodes, which are designed to emit energy about their entire circumference. Such current flow is akin to electrode arrangement shown in FIG. 5B of the '675 patent, which is not described as lateral energy delivery but rather is described as “uniform” energy delivery.

Ex. 2005 ¶ 76.

Patent Owner also contends that

Petitioner’s citation to FIG. 6D of the '675 patent also misses the mark. (Reply, 25.) In contrast to Imran, which includes numerous electrodes that continuously engage “the wall of the heart” (Ex-

1068, 2:44-45, 6:53-67), the FIG. 6D embodiment includes only two surface penetrating electrodes. (Ex-1001, 17:27-35.)

PO Sur-reply 22–23.

Thus, Patent Owner contends that Makower does not deliver energy laterally because, “[l]ike Imran, the uninsulated surfaces on Makower’s ring electrodes emit energy in all directions (uniformly per the parlance of the ’675 patent) and further result in current being drawn away from the target tissue.” PO Sur-reply 23.

e) Analysis

We find that Makower discloses electrosurgical devices for the nasal cavity or nasopharynx that use multiple bipolar electrodes that cauterize or heat adjacent tissue such that the function of a nasopharyngeal passage of the patient is altered. Ex. 1005 ¶¶ 43, 160–162, 164, 187, Fig. 5H; Pet. 28; *see supra* Section III.F.1.b.1. (discussing the operability of Makower’s device). We further agree with Petitioner that, in order to treat “tissue adjacent” to Makower’s electrodes, energy flows from the electrodes into tissue that is lateral relative to the distal portion of the device, as required by claim 22. Here, we credit the testimony of Dr. Bu-Saba that “a person of ordinary skill in the art would have understood that current traveling between Makower’s bipolar electrodes would create electrical fields that concentrate current lateral relative to the axis of the distal portion of the RF ablation probe and into the targeted tissue.” Ex. 1002 ¶ 206.

Patent Owner’s contentions to the contrary are unavailing. We considered Patent Owner’s arguments that Makower’s electrodes are configured to deliver energy uniformly as compared to laterally. PO Resp. 65 (citing Ex. 2003 ¶ 109; Ex. 2005 ¶¶ 74–75). However, we are persuaded by Petitioner’s arguments and the testimony of Dr. Bu-Saba that Makower’s

device directs energy laterally into the adjacent tissue that is in contact with the electrodes and would “not travel through surrounding air.” Ex. 1098 ¶ 152; *see also id.* (“As a matter of operation, electrodes of the device taught by Makower would contact target tissue and portions of the electrodes that are not contacting tissue would be insulated by air.”); *id.* (“a person of ordinary skill in the art would have understood that mucus or blood would not affect treatment or would be removed from the nasal cavity and nasopharynx prior to ablation”). Thus, even if the ring electrodes of Makower are capable of generally directing energy uniformly, in the context of Makower’s disclosed use, we find that “a POSITA would have expected ring electrodes in Makower [to] concentrate energy laterally because air would insulate any portion not contacting tissue.” Pet. Reply 24 (citing Ex. 1091, 218:9–11, 221:18–222:7; Ex. 1001, 18:11–14; 23:65–24:3; Ex. 1098 ¶¶ 147–148, 151–152).²⁰

For the reasons above, we determine that the preponderance of the evidence favors Petitioner’s challenge and that claim 22 would have been obvious over Makower and Fang.

²⁰ Patent Owner’s expert also admits that the ’675 patent’s Figure 5B, which purports to show a pair of bipolar ring electrodes, is able to deliver energy “laterally and uniformly.” Ex. 1089, 194:5–196:6. Patent Owner’s suggestion that the surface electrodes of the ’675 patent’s Figure 6D (which are shown creating a lateral energy field) are functionally distinct from surface electrodes, such as shown in Imran’s Figure 1, also fails. PO Sur-Reply 22–23 (arguing Figure 6D “includes only two *surface penetrating* electrodes”). The patent’s description of Figure 6D states that “electrode elements 72 and 74 . . . are adapted to extend laterally from the main catheter body, and *in some cases*, penetrate the surrounding tissue prior to application of energy.” Ex. 1001, 17:27–30 (emphasis added). Thus, the disclosure permits but does not require surface-penetrating electrodes.

4. *Other Claims*

Petitioner argues that dependent claims 2–3, 7, 9, 10, 18, 19, 20, 21, and 23–25 would have been obvious over Makower and Fang. Pet. 77–85, 87–90. In support, Petitioner cites the prior art’s teachings and declarant testimony. *Id.* (citing, e.g., Ex. 1005 ¶¶ 164, 167, 187, 192, Figs. 5B, 5H; Ex. 1006, 877–878; Ex. 1008, 9:14–45; Ex. 1002 ¶¶ 183–214). We have considered and credit this evidence and testimony, which we find establishes sufficiently that the combination of references teaches or suggests the limitations added by the dependent claims and that the POSA had reasons for combining the art as proposed to arrive at the claimed subject matter with a reasonable expectation of success.

Patent Owner does not argue the patentability of these dependent claims separately, relying instead on its general arguments against Ground 4 and claim 1, which we addressed and found unavailing as discussed above in Section III.F.1. *See* PO Resp. 30–31 (“Petitioners failed to establish that Makower’s device would have reasonably been expected to successfully achieve treatment of targeted tissue, as required by all claims.”).

Based on the preponderance of the evidence, we conclude that claims 2–3, 7, 9, 10, 18, 19, 20, 21, and 23–25 are unpatentable as obvious over Makower and Fang.

G. *Other Grounds: Grounds 1–3*

Petitioner argues that claims 1–3, 7, 9, 10, and 17–25 are anticipated by Saadat (Ground 1), that Saadat renders obvious claims 24 and 25 (Ground 2), and that Deem anticipates claims 1–3, 7, 9, 10, and 17–25 (Ground 3). Pet. 32–64. Each of Grounds 1–3 is premised on the challenged claims

being entitled to priority no earlier than October 22, 2019. *See, e.g.*, Pet. 2, 6–7, 54.

Petitioner met its burden to show unpatentability of all challenged claims under Ground 4 (as discussed above); we exercise our discretion not to reach Grounds 1–3. *Bos. Sci. Scimed, Inc. v. Cook Grp. Inc.*, 809 F. App’x 984, 990 (Fed. Cir. 2020) (nonprecedential) (explaining that “the Board has discretion to decline to decide additional instituted grounds once petitioner has prevailed on all its challenged claims.”).

IV. PATENT OWNER’S MOTION TO EXCLUDE

Patent Owner moves to exclude Exhibit 1113 and four paragraphs of Dr. Bu-Saba’s Reply Declaration discussing that exhibit. Paper 35, 1–5 (moving to exclude Exhibit 1113 and paragraphs 112–115 of Exhibit 1098). The gist of Patent Owner’s motion is that Exhibit 1113 comprises “new evidence” and Petitioner’s “belated submission” of that evidence is “highly prejudicial” to Patent Owner and, therefore, Exhibit 1113 (and the testimony about it) should be excluded under Fed. R. Evid. 403. *Id.* Petitioner filed an opposition to the motion (Paper 36), which was followed by Patent Owner’s reply to that opposition (Paper 37).

Our Decision does not rely on any of the challenged evidence, and we therefore dismiss Patent Owner’s Motion to Exclude as moot.

V. CONCLUSION²¹

Petitioner has shown by a preponderance of the evidence that the challenged claims are unpatentable as summarized below.

Claims	35 U.S.C. §	Reference(s)/Basis	Claims Shown Unpatentable	Claims Not shown Unpatentable
1–3, 7, 9, 10, 17–25	102	Saadat ²²		
24, 25	103	Saadat ²³		
1–3, 7, 9, 10, 17–25	102	Deem ²⁴		
1–3, 7, 9, 10, 17–25	103	Makower, Fang, Edwards	1–3, 7, 9, 10, 17–25	
Overall Outcome			1–3, 7, 9, 10, 17–25	

VI. ORDER

In consideration of the foregoing, it is hereby:

²¹ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. § 42.8(a)(3), (b)(2).

²² Because all challenged claims are determined to be unpatentable based on other asserted grounds, we decline to further address this additional ground.

²³ Because all challenged claims are determined to be unpatentable based on other asserted grounds, we decline to further address this additional ground.

²⁴ Because all challenged claims are determined to be unpatentable based on other asserted grounds, we decline to further address this additional ground.

ORDERED that Petitioner has proved by a preponderance of the evidence that claims 1–3, 7, 9, 10, and 17–25 of U.S. Patent No. 10,610,675 B2 are unpatentable;

FURTHER ORDERED that Patent Owner’s Motion to Exclude (Paper 35) is *dismissed* as moot; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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Patent 10,610,675 B2

For PETITIONER:

David Cavanaugh
Alexis Cohen
Alexander Nemtzow
Gilbert Smolenski
WILMER CUTLER PICKERING HALE AND DORR LLP
david.cavanaugh@wilmerhale.com
alexis.cohen@wilmerhale.com
alex.nemtzw@wilmerhale.com
gilbert.smolenski@wilmerhale.com

FOR PATENT OWNER:

Heath Briggs
Trenton Ward
Elana Araj
Benjamin Gilford
GREENBERG TRAURIG, LLP
briggsh@gtlaw.com
trenton.ward@gtlaw.com
araje@gtlaw.com
gilfordb@gtlaw.com

Jason Stach
Sonja Sahlsten
Anthony Berlenbach
Kathleen Daley
Christopher Howes
FINNEGAN, HENDERSON, FARABOW,
GARRETT, & DUNNER LLP
jason.stach@finnegan.com
sonja.sahlsten@finnegan.com
anthony.berlenbach@finnegan.com
kathleen.daley@finnegan.com
christopher.howes@finnegan.com